

SkinCheck AI



By:

Ali Hassan

20036

Abaid Daud

20453

Supervised by:

Hasnat Ali

Lecturer

Faculty of Computing
Riphah International University, Islamabad
Spring/Fall 2023

A Dissertation Submitted To

Faculty of Computing,

Riphah International University, Islamabad

As a Partial Fulfillment of the Requirement for the Award of the

Degree of

Bachelors of Science in Computer Science

Faculty of Computing

Riphah International University, Islamabad

Date: [date of final presentation]

Final Approval

This is to certify that we have read the report submitted by *Ali Hassan (Sap Id #20036)* and *Abaid Daud (Sap Id #20453)* for the partial fulfillment of the requirements for the degree of the Bachelors of Science in Computer Science (BSCS). It is our judgment that this report is of sufficient standard to warrant its acceptance by Riphah International University, Islamabad for the degree of Bachelors of Science in Computer Science (BSCS).

Committee:

1

Mr. Hasnat Ali
(Supervisor)

2

Dr. Musharaf Ahmed
(Head of Department/chairman)

Declaration

We hereby declare that this document “**Skin Disease Detection using Deep Learning (SkinCheck AI)**” neither as a whole nor as a part has been copied out from any source. It is further declared that we have done this project with the accompanied report entirely on the basis of our personal efforts, under the proficient guidance of our teachers, especially our supervisor **Mr. Hasnat Ali**. If any part of the system is proved to be copied out from any source or found to be reproduction of any project from anywhere else, we shall stand by the consequences.

Ali Hassan

SAP ID # 20036

Abaid Daud

SAP ID # 20453

Dedication

We would like to dedicate this project report to three important people who have played a crucial role in our academic and personal journey.

Firstly, we dedicate this report to our loving parents, whose unwavering support and encouragement drive our achievements. Their sacrifices and belief in us have allowed us to pursue our dreams and reach this milestone. We are forever grateful for their love, guidance, and inspiration, and we hope to make them proud with our future endeavors.

Secondly, we would like to express our heartfelt appreciation to our supervisor Mr. Hasnat Ali, whose guidance and expertise have been invaluable throughout the project. Your constant encouragement, feedback, and constructive criticism have helped us refine our ideas and develop our research skills. We are grateful for the opportunity to work under your guidance, and we have learned a great deal from your mentorship.

To our dearest mother, father, and supervisor, thank you for being our support pillars and always believing in us. This report is a testament to the efforts and contributions of each of you, and we are honored to dedicate it to you.

Acknowledgment

First of all, we are obliged to Allah Almighty, the Merciful, the Beneficent, and the source of all knowledge, for granting us the courage and knowledge to complete this project.

We would like to express our sincere gratitude to everyone who has contributed to the completion of this project. Your support, guidance, and encouragement have been invaluable, and we could not have done it without you.

First and foremost, we would like to thank my supervisor, Mr. Hasnat Ali, for his unwavering support and guidance throughout the project. Mr. Hasnat Ali has been instrumental in helping us develop our ideas and implement them effectively. His expert knowledge, critical feedback, and constructive criticism have been instrumental in refining the project to its current state. We are fortunate to have had the opportunity to work under his guidance.

We would also like to acknowledge Mr. Shahbaz Hassan for his valuable contributions to the project. Mr. Shahbaz Hassan's expertise in AI case study building and other approaches has been a great asset in the development of the project. His insightful feedback and suggestions have been instrumental in shaping the project and adding value to it.

We are also grateful to my family for their constant support and encouragement throughout this project. Our parents, in particular, have been our pillars of strength, and their love and belief in us have motivated me to strive for excellence.

Finally, we would like to thank all my classmates and friends who have provided their support and encouragement.

Ali Hassan
SAP ID # 20036

Abaid Daud
SAP ID # 20453

Table of Contents

Table of Contents	i
List of Tables	iii
List of Figures	iv
Abstract	5
Chapter 1: Introduction	6
1.1 Goals and Objectives	6
1.2 Scope of the Project	7
Chapter 2: Literature Review	10
2.1 Introduction	10
2.2 Background and Problem Elaboration	10
2.3 Detailed Literature Review	11
2.3.1 Definitions	11
2.3.2 Related Research Work 1	12
2.3.3 Related Research Work 2	13
2.3.4 Related Research Work 3	14
2.3.5 Related Research Work 4	15
2.4 Literature Review Summary Table	16
2.5 Market Survey	17
2.5.1 Survey 1	17
2.5.2 Survey 2	18
2.5.3 Survey 3	19
2.6 Market Survey Graph	20
2.7 Research Gap	20
2.8 Problem Statement	21
Chapter 3: Requirements and Design	23
3.1 Requirements	23
3.1.1 Functional Requirements	23
3.1.2 Non-Functional Requirements	24
3.1.3 Hardware and Software Requirements	24
3.2 Proposed Methodology	25
3.3 System Architecture	26
3.4 Use Cases	27
3.4.1 Patient Registration and Login	27
3.4.2 Profile Management	27
3.4.3 Search and Book Appointments	28
3.4.4 Doctor Profile Management	29
3.4.5 Doctor Registration and Profile Creation	29
3.4.6 Online Consultations	30
3.4.7 Appointment Management	30
3.4.8 Skin Disease Detection	31
3.4.9 Payment Processing	31
3.4.10 Withdrawal of Earnings	32
3.4.11 Rate and Review	33
3.4.12 User Management	33
3.4.13 Doctor Verification	34
3.4.14 Appointment Management	34
3.5 Methodology Diagram	36
3.6 Sequence diagram	37
3.7 Use Case Diagram	38

3.8 CNN Architecture Diagram	39
3.9 GUI Graphical User Interfaces (<i>Optional</i>)	39
3.9.1 Home Page	39
3.9.2 Login	41
3.9.3 SignUp	42
3.9.4 Patient Dashboard	43
3.9.5 Doctors Dashboard.....	50
Chapter 4: Implementation and Test Cases	58
4.1 Implementation	58
4.1.1 Implementation of First Component/Algorithm	58
4.2 Test case Design and description.....	59
4.2.1 Sample Test case No.1	60
4.2.2 Sample Test case No.2	60
4.3 Test Metrics	60
4.3.1 Sample Test case Matric.No.1	60
4.3.2 Sample Test case Metric.No.2	61
Chapter 5: Experimental Results and Analysis.....	62
Chapter 6: Conclusion and Future Directions.....	63
6.1 Introduction.....	63
6.2 Conclusions and Summary of Work Done	63
6.3 Coverage of Project Scope and Objectives.....	63
6.4 Challenges Faced	64
6.5 Remaining Work and Future Directions	64
6.6 Recommendations for Future Enhancements	64
6.7 Proposed Plan for FYP-Part 2.....	65
References	66

List of Tables

Table 1: History of Skin Disease Detection Models.....	16
Table 2: Patient Registration and Login	27
Table 3: Profile Management.....	27
Table 4: Search and Book Appointments	28
Table 5: Doctor Profile Management	29
Table 6: Doctor Registration and Profile Creation	29
Table 7: Online Consultations	30
Table 8: Appointment Management	30
Table 9: Skin Disease Detection	31
Table 10: Payment Processing	31
Table 11: Withdrawal of Earnings.....	32
Table 12: Rate and Review	33
Table 13: User Management.....	33
Table 14: Doctor Verification.....	34
Table 15: Appointment Management	34

List of Figures

Figure 1 Marketing Survey	20
Figure 2: Methodology Diagram	36
Figure 3: Sequence Diagram.....	37
Figure 4 Use Case Diagram	38
Figure 5: CNN Architecture Diagram.....	39

Abstract

Skin diseases, affecting millions worldwide, often present challenges in diagnosis due to their similar visual appearance and the unwillingness of patients to seek medical advice. This report presents SkinCheck AI, an innovative application designed to support dermatologists in diagnosing skin diseases through Convolutional Neural Networks (CNNs) and image-based deep learning classification. The app provides a reliable second opinion on diagnoses by analyzing user-uploaded images. Additionally, the app offers features to facilitate patient-doctor interactions, such as video consultations and sharing resources on skin conditions and treatments. Data augmentation techniques are employed to enhance the accuracy of disease detection. The development and implementation of SkinCheck AI aim to improve early detection, simplify the consultation process, and ultimately lead to better treatment outcomes and increased patient satisfaction.

The proposed solution has several advantages, including the ability of CNNs to recognize and classify skin diseases and a user-friendly application that takes an image as input and outputs the predicted disease type. The app will provide a virtual consultation feature for dermatologists to diagnose patients remotely, especially for those who cannot easily access in-person care. Additionally, the app will offer educational resources and information on skin conditions, treatments, and preventive measures to increase awareness and adherence to treatment plans. In conclusion, the SkinCheck AI app has the potential to revolutionize healthcare by providing better and more accessible diagnoses of skin diseases through the use of deep learning algorithms, ultimately improving the quality of life for many people around the world.

Chapter 1: Introduction

Skin diseases pose significant challenges in diagnosis and treatment, affecting millions worldwide. The visual similarities among various skin conditions and a patient's reluctance to seek medical advice often lead to diagnostic uncertainties and delayed treatment. In response to this pressing issue, this paper introduces SkinCheck AI. This innovative application leverages Convolutional Neural Networks (CNNs) and image-based deep-learning classification to support dermatologists in diagnosing skin diseases better.

The primary objective of SkinCheck AI is to provide a reliable second opinion for dermatologists on skin disease diagnoses by analyzing user-uploaded images. Through the power of deep learning algorithms, the application aims to reduce diagnostic uncertainties and enhance the accuracy of disease detection. By utilizing CNNs, SkinCheck AI can effectively recognize and classify different skin diseases, offering dermatologists an invaluable tool for their diagnostic decision-making process.

Beyond its diagnostic capabilities, SkinCheck AI facilitates patient-doctor interactions and streamers the consultation process. The application offers features such as video consultations and sharing resources on various skin conditions and treatments. These features enhance the convenience of accessing dermatological care and encourage patients to seek medical advice and adhere to treatment plans.

The implementation of SkinCheck AI has the potential to revolutionize healthcare by addressing the challenges associated with skin disease diagnosis. Its user-friendly interface, which takes an image as input and provides predicted disease types as output, ensures accessibility for dermatologists. Additionally, the app's virtual consultation feature enables remote diagnosis, making it particularly beneficial for individuals facing barriers to in-person care.

Furthermore, SkinCheck AI aims to increase awareness and knowledge about skin conditions by offering educational resources and information on treatments and preventive measures. By empowering individuals with comprehensive information, the application promotes the proactive management of skin diseases and encourages early detection.

1.1 Goals and Objectives

The development and implementation of the SkinCheck AI application are driven by specific goals and objectives aimed at addressing the challenges associated with skin disease diagnosis

Introduction

and improving the overall healthcare experience for both patients and dermatologists. The following are the key goals and objectives of SkinCheck AI:

- **Improve Diagnostic Accuracy:** The primary goal of SkinCheck AI is to enhance the accuracy of skin disease diagnosis. By utilizing Convolutional Neural Networks (CNNs) and image-based deep learning classification, the application aims to provide dermatologists with a reliable second opinion on diagnoses. This objective seeks to reduce diagnostic uncertainties and ensure better identification and classification of various skin conditions.
- **Enhance Patient-Doctor Interactions:** SkinCheck AI facilitates seamless and effective interactions between patients and dermatologists. The application seeks to streamline the consultation process through features like video consultations enabling patients to access dermatological care conveniently. This objective focuses on improving patient satisfaction and engagement while ensuring timely and personalized healthcare services.
- **Promote Accessibility to Dermatological Care:** The application addresses barriers to accessing dermatological care by providing a virtual consultation feature. This objective is significant for individuals facing geographical or logistical challenges seeking in-person care. SkinCheck AI aims to increase accessibility and ensure that individuals receive timely and better diagnoses, regardless of location.
- **Provide Educational Resources** SkinCheck AI goes beyond diagnosis and treatment by offering educational resources and information on skin conditions, treatments, and preventive measures. This objective aims to increase awareness and knowledge among users, empowering them to actively participate in managing their skin health. By providing comprehensive information, the application seeks to promote proactive measures.
- **Enhance Treatment Outcomes and Patient Satisfaction:** The ultimate objective of SkinCheck AI is to improve treatment outcomes and patient satisfaction. The application aims to initiate timely treatments and interventions by facilitating better and early diagnoses, leading to better disease management. This objective focuses on enhancing patient satisfaction and overall quality of life by ensuring effective and personalized care.

1.2 Scope of the Project

The scope of the SkinCheck AI project encompasses the development, implementation, and deployment of an innovative application designed to support dermatologists in diagnosing skin

diseases and enhancing the overall healthcare experience for patients. The project involves the following key components:

- **Skin Disease Diagnosis:** The project utilizes deep learning algorithms, particularly CNNs, for skin disease recognition and classification. These algorithms will be trained using a dataset of skin condition images to ensure accurate disease detection. Data augmentation techniques will be employed to enhance the performance of the algorithms.
- **Second Opinion Support:** SkinCheck AI will be a valuable tool for dermatologists by providing a reliable second opinion on skin disease diagnoses. The system will analyze the image uploaded by dermatologists and predict the disease name to reduce confusion in the decision-making process.
- **Remote Diagnosis and Accessibility:** The project will focus on enabling remote diagnosis of skin diseases through the SkinCheck AI application. This feature will particularly benefit individuals facing challenges accessing in-person dermatological care. By offering virtual consultations, the application aims to increase accessibility to dermatological services and ensure timely diagnoses for patients regardless of geographical location.
- **Educational Resources:** SkinCheck AI will include a comprehensive repository of educational resources on various skin conditions, treatments, and preventive measures. The application will provide accessible and reliable information to empower users to understand their skin health better, actively participate in disease management, and promote early detection and prevention.
- **User Interaction Features:** The SkinCheck AI application will incorporate features to facilitate seamless interactions between patients and dermatologists. These features may include video consultation capabilities and secure information sharing between patients and healthcare professionals. The application will ensure ease of use for both patients and dermatologists.

The scope of the SkinCheck AI project is comprehensive, covering various aspects related to the application's development, implementation, and utilization. It includes the development of a user-friendly application that utilizes Convolutional Neural Networks (CNNs) and image-based deep-learning classification techniques to support dermatologists in diagnosing skin diseases better. Additionally, the application encompasses features such as video consultations, appointment booking, and information sharing on skin conditions and treatments, aiming to streamline patient-doctor interactions and improve the overall healthcare experience.

Introduction

Furthermore, the project emphasizes the importance of accessibility by providing virtual consultation capabilities, enabling remote diagnosis for individuals who face challenges in accessing in-person care.

The project focuses on developing and implementing the SkinCheck AI application for skin disease diagnosis and patient-doctor interactions. The project does not encompass the provision of immediate medical treatment or replace the expertise of dermatologists. The application is intended to support healthcare professionals in their diagnostic processes and enhance the patient's healthcare journey.

Chapter 2: Literature Review

2.1 Introduction

This chapter serves as a comprehensive review of the existing literature on skin disease detection and diagnosis. It begins by introducing the background and context of the challenges associated with accurate diagnosis in dermatology. The review encompasses an in-depth analysis of relevant research studies, algorithms, and methodologies employed by previous applications in this domain.

The literature review examines the definitions and descriptions of key concepts and approaches for skin disease detection. It explores the various techniques and methods employed by different inventors and researchers. Each reviewed work is accompanied by its corresponding name, reference, inventor, and year of publication.

A summary table facilitates a comparative analysis of the reviewed works. The table presents a systematic overview of the reviewed literature, including the input and output of each approach, as well as a concise description of the method and its findings.

Furthermore, this chapter identifies the research gap and highlights the problem statement the current project aims to address. This sets the foundation for the subsequent chapters, which will delve into the development and implementation of SkinCheck AI, leveraging the insights gained from the literature review.

Overall, this literature review establishes a solid understanding of the existing knowledge and research landscape in skin disease detection and diagnosis, providing a valuable framework for developing and evaluating SkinCheck AI.

2.2 Background and Problem Elaboration

In Background and Problem Elaboration, we provide a detailed background of the challenges associated with accurately diagnosing skin diseases and further elaborate on the problem SkinCheck AI aims to address.

Skin diseases present a significant burden on individuals and healthcare systems worldwide. The visual similarities among various skin conditions and the complexity of dermatological diagnoses often lead to misdiagnosis or delayed treatment. Moreover, the limited accessibility to dermatologists in certain areas exacerbates the problem, preventing timely and accurate diagnosis for many patients.

Traditional methods of diagnosing skin diseases rely heavily on visual inspection by dermatologists, which is subjective and prone to human error. This subjective nature and the

increasing number of skin disease cases highlight the need for an automated system to assist dermatologists in accurate diagnosis and improve overall patient care.

The emergence of artificial intelligence (AI) and deep learning techniques has shown great potential in various medical applications, including skin disease detection. By leveraging advanced machine learning algorithms, such as Convolutional Neural Networks (CNNs), it is possible to develop a system that can analyze images of skin conditions and provide reliable diagnostic support.

SkinCheck AI aims to bridge the gap in skin disease diagnosis by utilizing deep learning algorithms and image-based classification to provide dermatologists with a reliable second opinion. By leveraging the power of AI, SkinCheck AI can enhance diagnostic accuracy, reduce uncertainty, and ultimately improve patient outcomes. Furthermore, SkinCheck AI can overcome the barriers of limited dermatologist access, making quality dermatological care more accessible to individuals, regardless of their geographical location.

2.3 Detailed Literature Review

2.3.1 Definitions

- **Skin Disease Recognition:** Skin Disease Recognition automatically identifies and categorizes various skin diseases based on visual patterns and features present in skin condition images. This project involves using advanced techniques, such as deep learning algorithms, particularly Convolutional Neural Networks (CNNs), to analyze skin condition images and classify them into different disease categories.
- **Deep Learning:** Deep Learning is a subfield of machine learning that focuses on training artificial neural networks to automatically learn and extract high-level representations from complex data. It involves constructing deep neural networks with multiple layers that can efficiently learn patterns and relationships in large datasets. Deep learning algorithms, such as CNNs, have succeeded in various domains, including image and speech recognition, natural language processing, and skin disease recognition. The Skin Disease Recognition Project aims to develop a robust and accurate system for diagnosing skin diseases by leveraging deep learning techniques.
- **Convolutional Neural Networks (CNNs):** Convolutional Neural Networks (CNNs) are a class of deep learning algorithms specifically designed for image analysis tasks. They are inspired by the visual processing mechanisms of the human brain and excel at capturing spatial relationships and extracting meaningful features from images. CNNs consist of multiple interconnected layers, including convolutional, pooling, and fully connected

layers. The convolutional layers apply filters to input images to detect relevant patterns, while the pooling layers down sample the feature maps to reduce the spatial dimensionality. CNNs have revolutionized the field of computer vision and are widely used for tasks like image classification, object detection, and in this project, skin disease recognition.

- **Data Augmentation:** Data Augmentation is a technique used to increase the diversity and quantity of a given dataset by applying various transformations and modifications to the existing data. In the Skin Disease Recognition Project context, data augmentation is employed to create additional training examples for the deep learning models. Common data augmentation techniques include rotation, scaling, flipping, cropping, and introducing noise to the images. By augmenting the dataset, the models become more robust to variations and improve generalization, ultimately enhancing their performance and reducing the risk of overfitting.
- **Algorithm:** An Algorithm is a well-defined set of instructions or rules that describe a step-by-step procedure to solve a specific problem. In the Skin Disease Recognition Project context, algorithms refer to the computational procedures implemented to analyze skin condition images and classify them into different disease categories. This includes pre-processing steps, feature extraction methods, and deep learning models' training and inference processes. Algorithms play a crucial role in automating the diagnosis of skin diseases and providing accurate predictions based on the given input data.
- **Machine Learning:** Machine Learning focuses on developing algorithms and techniques that enable computers to learn from data and make predictions or decisions without being explicitly programmed. It encompasses various approaches, including supervised, unsupervised, and reinforcement learning. In the Skin Disease Recognition Project context, machine learning techniques, particularly deep learning, are utilized for training models to recognize and classify skin diseases based on patterns and features learned from a labelled dataset. Machine learning enables the system to improve its performance over time through iterative learning from examples.

2.3.2 Related Research Work 1

The research work paper titled "Computer Vision Based Skin Disorder Recognition using EfficientNet: A Transfer Learning Approach" presents a novel approach to recognizing skin disorders using computer vision. The authors have used EfficientNet, a state-of-the-art CNN architecture consisting of 8 models between B0 and B7. In this study, all models of the

EfficientNet family have been used with the transfer learning approach to recognize 20 different types of skin disorders and one healthy skin with better classification accuracy.

The authors have achieved an accuracy of 97.10% using EfficientNet-B7, which is better than other models of EfficientNet.

The methodology adopted in this study involves collecting data from publicly available datasets such as ISIC-2019 and HAM10000. The collected data has been preprocessed by resizing the images to 224x224 pixels and normalizing them. The authors have used transfer learning to fine-tune the pre-trained EfficientNet models on their dataset.

Despite achieving high accuracy, this research work has some limitations. One limitation is that the dataset used in this study only contains images from two sources, which may only represent some skin disorders. Another limitation is that the dataset only contains images from lighter-skinned individuals, which may not generalize well to darker-skinned individuals.

In conclusion, this paper presents a novel approach to recognizing skin disorders using computer vision with high accuracy. The authors have used the EfficientNet-B7 model to recognize 20 different types of skin disorders and one healthy skin with better classification accuracy. However, some limitations in this research work need to be addressed in future studies. This research work can be useful for dermatologists and healthcare professionals for early detection and diagnosis of various skin disorders.

2.3.3 Related Research Work 2

This paper titled "A Skin Disease Classification Model Based on DenseNet and ConvNeXt Fusion" by Wei et al. proposes a convolutional neural network (CNN) model for skin disease classification based on model fusion. The authors chose DenseNet201 and ConvNeXt_L as the backbone sub-classification models of their model fusion. They introduced an attention module on the core block of each sub-classification model to assist the network in acquiring a region of interest to enhance the ability of the network model to extract image features. The features extracted by the shallow network could capture more details, while the features extracted by the deep network contained more abstract semantic information. Combining these characteristics, a parallel strategy was adopted to fuse the features of the deep and shallow layers.

The authors evaluated their proposed model on three datasets: ISIC 2018, HAM10000, and PH2. They achieved an accuracy of 95.5% on ISIC 2018 and 92.3% on HAM10000 outperforming existing state-of-the-art methods.

The significance of using DenseNet and ConvNeXt Fusion in skin disease classification lies in their ability to extract high-level features from images with fewer parameters than traditional CNNs. This results in faster training times and better performance compared to other models. However, there are some limitations to this study. One limitation is that it only focuses on three datasets, which may not be representative of all skin diseases or all populations. Another limitation is that it does not consider other factors such as age or gender that may affect skin disease diagnosis.

In terms of methodology, Wei et al. used transfer learning to fine-tune pre-trained models for skin disease classification. They also used data augmentation techniques such as rotation, flipping, and scaling to increase the size of their datasets.

Overall, this paper presents a comprehensive study on skin disease classification using advanced deep learning techniques. The proposed model achieves high accuracy and outperforms existing state-of-the-art methods. However, there are limitations to this study that should be considered when applying these findings to real-world scenarios.

The authors also highlight some of the challenges and limitations in skin disease classification. One of the main challenges is the large variability in skin lesions, which can make it difficult to accurately diagnose certain diseases. Additionally, there is a lack of standardization in image acquisition and annotation, which can lead to inconsistencies in datasets and affect the performance of machine learning models.

To address these challenges, Wei et al. propose using a model fusion approach that combines the strengths of different CNN architectures to improve accuracy and generalization ability. They also suggest that future research should focus on developing standardized protocols for image acquisition and annotation to improve the quality and consistency of datasets.

In conclusion, this paper presents a valuable contribution to the field of skin disease classification using advanced deep learning techniques. The proposed model achieves high accuracy and outperforms existing state-of-the-art methods. However, there are limitations to this study that should be considered when applying these findings to real-world scenarios. Future research should focus on addressing these limitations and developing standardized protocols for image acquisition and annotation to improve the quality and consistency of datasets.

2.3.4 Related Research Work 3

This research review paper titled "Opportunities and Challenges: Classification of Skin Disease Based on Deep Learning" by Bin Zhang et al. discusses the use of deep learning in skin disease

Literature Review

classification. The authors provide a comprehensive overview of the opportunities and challenges in using deep learning for skin disease diagnosis.

The paper discusses various technical terms related to deep learning, such as convolutional neural networks (CNNs), transfer learning, and data augmentation. The authors also mention the model they worked on, which is a CNN-based model called Inception-v3.

The authors focused on predicting various skin diseases such as melanoma, basal cell carcinoma, and squamous cell carcinoma. They achieved high accuracy rates ranging from 85% to 90% for different skin diseases using their proposed model.

However, the authors also highlight several problems and limitations they faced during their research work. One major limitation is the lack of large-scale annotated datasets for skin disease classification. Another limitation is the difficulty in obtaining accurate ground truth labels for skin lesions due to inter-observer variability among dermatologists.

To overcome these limitations, the authors adopted a methodology that involved collecting and preprocessing large-scale datasets from various sources. They also used transfer learning techniques to fine-tune pre-trained models on their specific dataset. Additionally, they used data augmentation techniques to increase the size of their dataset and improve model generalization.

In conclusion this literature review provides valuable insights into the opportunities and challenges of using deep learning for skin disease classification. The authors highlight several limitations that need to be addressed in future research work to improve the accuracy and generalization of deep learning models for skin disease diagnosis.

2.3.5 Related Research Work 4

The paper "Deep Learning Based Automated Diagnosis of Skin Diseases Using Dermoscopy" by Vatsala Anand, Sheifali Gupta, Deepika Koundal, Shubham Mahajan, Amit Kant Pandit, and Atef Zaguia proposes an automated deep learning model for diagnosing skin diseases at an early stage using dermoscopy images. The authors used the HAM10000 dataset to train and validate the model.

The authors achieved an accuracy of 95.5% in classifying seven different types of skin diseases: melanoma, nevus, seborrheic keratosis, basal cell carcinoma, actinic keratosis/bowen's disease (intraepithelial carcinoma), dermatofibroma and vascular lesions. The proposed model has shown promising results which will act as a second opinion tool for dermatologists.

The authors reported that they achieved an accuracy of 95% on ResNet50, ResNet101 achieved an accuracy of 89%, and ResNet18 achieved an accuracy of 86%. The proposed model also

outperformed all three versions of the ResNet model in terms of sensitivity, specificity, precision, and F1 score.

However, there are some limitations to this study. One limitation is that the dataset used in this study only includes images from a single source (HAM10000). This may limit the generalizability of the proposed model to other datasets or real-world clinical settings. Another limitation is that the proposed model does not consider other factors such as patient history or other clinical information that may be relevant for accurate diagnosis.

The methodology adopted by the authors involves preprocessing the dermoscopy images by resizing them to 224x224 pixels and normalizing them using mean and standard deviation values calculated from the training set. The authors then trained their deep learning model using transfer learning with pre-trained weights from VGG16 architecture.

In conclusion, this paper presents a promising approach for automated diagnosis of skin diseases using dermoscopy images. The proposed model achieved high accuracy in classifying seven different types of skin diseases. However, there are some limitations to this study that need to be addressed in future research. Overall, this work has the potential to revolutionize the field of dermatology by providing an automated tool for accurate and efficient diagnosis of skin diseases.

2.4 Literature Review Summary Table

Table 1: History of Skin Disease Detection Models

No.	Name, reference	Year	Input	Output	Description
1.	Computer Vision Based Skin Disorder Recognition using EfficientNet: A Transfer Learning Approach, [1]	2021	Skin Disease Image	Diagnoses of 20 skin diseases	EfficientNet-B7 model to recognize 20 different types of skin disorders and one healthy skin condition with a classification accuracy of 97.10%. Limitations include the use of a dataset from only two sources, potentially lacking representation of all skin disorders, and the exclusion of images from individuals with darker skin tones.

2.	A Skin Disease Classification Model Based on DenseNet and ConvNeXt Fusion, [2]	2023	Skin Disease Image	Diagnoses of 7 skin diseases	Classification model that combines DenseNet201 and ConvNeXt_L architectures. Accuracy achieved 92 to 95%. Limitations include large variability in skin lesions, lack of standardization in image acquisition and annotation, which can lead to inconsistencies.
3.	Opportunities and Challenges: Classification of Skin Disease Based on Deep Learning, [3]	2021	Skin Disease Image	Diagnoses of 4 skin diseases	Inception V-3 model to diagnose 4 skin disorders with an accuracy of 85 to 90%. Limitations include lack of large-scale annotated datasets, difficulty in obtaining accurate ground truth labels for skin lesions.
4.	Deep Learning Based Automated Diagnosis of Skin Diseases Using Dermoscopy, [4]	2022	Skin Disease Image	Diagnoses of 7 skin diseases	ResNet50 achieved an accuracy of 95%, ResNet101 89%, and ResNet18 achieved an accuracy of 86%. Limitation is that the dataset used in this study only includes images from a single source (HAM10000).

2.5 Market Survey

2.5.1 Survey 1

The website named SkinCheck uses a deep learning algorithm to analyze skin images and detect skin cancer. The algorithm was developed by a team of researchers at the University of California, San Francisco and was launched in 2014.

It is trained on a large dataset of skin images, including both cancerous and non-cancerous lesions. The algorithm is able to identify skin cancer with a high degree of accuracy.

There are a few problems with the SkinCheck website. First, it is not available in all countries. Second, it is not always easy to use. Third, the accuracy of the algorithm is not always perfect.

SkinCheck AI

SkinCheck can diagnose a variety of skin diseases, including melanoma, basal cell carcinoma, and squamous cell carcinoma. It can also diagnose other skin conditions, such as actinic keratosis and psoriasis.

Overall, SkinCheck is a promising tool for early detection of skin cancer. However, it is important to note that it is not a replacement for a clinical exam by a dermatologist. If you have any concerns about your skin, you should see a doctor.

The algorithm is based on a convolutional neural network (CNN). CNNs are a type of deep learning algorithm that are well-suited for image classification tasks.

The algorithm was trained on a dataset of over 100,000 skin images. The dataset includes both cancerous and non-cancerous lesions.

The algorithm has been shown to be able to identify skin cancer with a high degree of accuracy. In one study, the algorithm was able to correctly identify skin cancer with an accuracy of 87%. Despite its high accuracy, the SkinCheck algorithm is not perfect. In some cases, the algorithm may incorrectly identify a non-cancerous lesion as cancerous. This is why it is important to see a doctor if you have any concerns about your skin.

SkinCheck is a valuable tool for early detection of skin cancer. However, it is important to use it in conjunction with other methods of skin cancer screening, such as regular skin exams by a dermatologist.

2.5.2 Survey 2

DermaCompare is a skin cancer screening platform that uses artificial intelligence, military-grade surveillance technology, crowdsourced data, and a cloud-based mobile app to help physicians and patients detect and monitor changes in their skin characteristics ¹²³.

DermaCompare was launched in 2016 by Emerald Medical Applications Corp., an Israeli-based company. It won the Publicis Groupe “Publicis90” Initiative Grand Prize at the inaugural Viva Technology Paris event in 2016, where it competed with 3,500 other tech companies. It also received CE certification in Europe and FDA clearance in the US.

DermaCompare uses an image-matching algorithm that compares the patient’s skin images with a database of images of diagnosed lesions. The algorithm assigns a risk score to each lesion based on its similarity to malignant or benign lesions in the database. The risk score ranges from 1 (low risk) to 5 (high risk) and is accompanied by a recommendation to consult a dermatologist or not ⁴.

The problem DermaCompare wants to solve is the early detection and prevention of skin cancer, especially melanoma, the deadliest form of skin cancer. Melanoma can be cured if

detected early, but it can also spread rapidly and become fatal if left untreated. Visual dermatologist inspection is the gold standard for diagnosis, but it can be subjective, costly, and inaccessible for many people. DermaCompare provides a low-cost, accessible, and objective tool for screening and monitoring skin lesions 23.

DermaCompare can diagnose various types of skin cancer, such as basal cell carcinoma, melanoma, actinic keratosis, and squamous cell carcinoma. It can detect benign lesions, such as nevi (moles), seborrheic keratosis, lentigo, and hemangioma 4.

2.5.3 Survey 3

SkinVision was launched in 2018 by SkinVision B.V., a Dutch-based company. It is a prominent online platform dedicated to assisting users in the early detection and monitoring of skin conditions, particularly skin cancer. Skin Vision employs advanced technology, including a sophisticated algorithm, to examine images of skin lesions and provide potential assessments and risk evaluations.

The algorithm utilized by Skin Vision is based on artificial intelligence and machine learning techniques. It incorporates a combination of computer vision algorithms and deep learning models to analyze key features of skin lesions, such as shape, color, texture, and border irregularities. By comparing these characteristics with a vast database of dermatological images, the algorithm can provide users with preliminary evaluations and indications of potential risks associated with their skin conditions.

Skin Vision's algorithm can diagnose several skin conditions, including various forms of skin cancer. It covers common types such as melanoma, basal cell carcinoma, squamous cell carcinoma, and other benign skin conditions like moles and keratoses. The ability to detect and monitor a wide range of skin conditions makes the Skin Vision website valuable for individuals seeking to proactively manage their skin health.

However, as with any technology-based platform, the Skin Vision website may encounter certain challenges and areas for improvement. One common issue users report is the occasional difficulty in accurately assessing certain skin lesions, particularly those with atypical or ambiguous characteristics. While the algorithm is continuously refined and optimized, the complexity and variability of skin conditions present ongoing challenges in achieving high diagnostic accuracy in all cases.

Understanding the technical aspects of the Skin Vision website, including the algorithm used, problem areas, and the number of diseases it diagnoses, is crucial for evaluating its effectiveness and identifying potential areas for improvement. By participating in this research

survey, you can provide valuable insights that contribute to this platform's ongoing development and enhancement, ultimately benefiting individuals in the early detection and management of skin conditions, particularly skin cancer.

2.6 Market Survey Graph

The graph displays the results of a marketing survey conducted for five years (2018-2022) on three different skin disease applications: SkinCheck, DermaCompare, and Skin Vision.

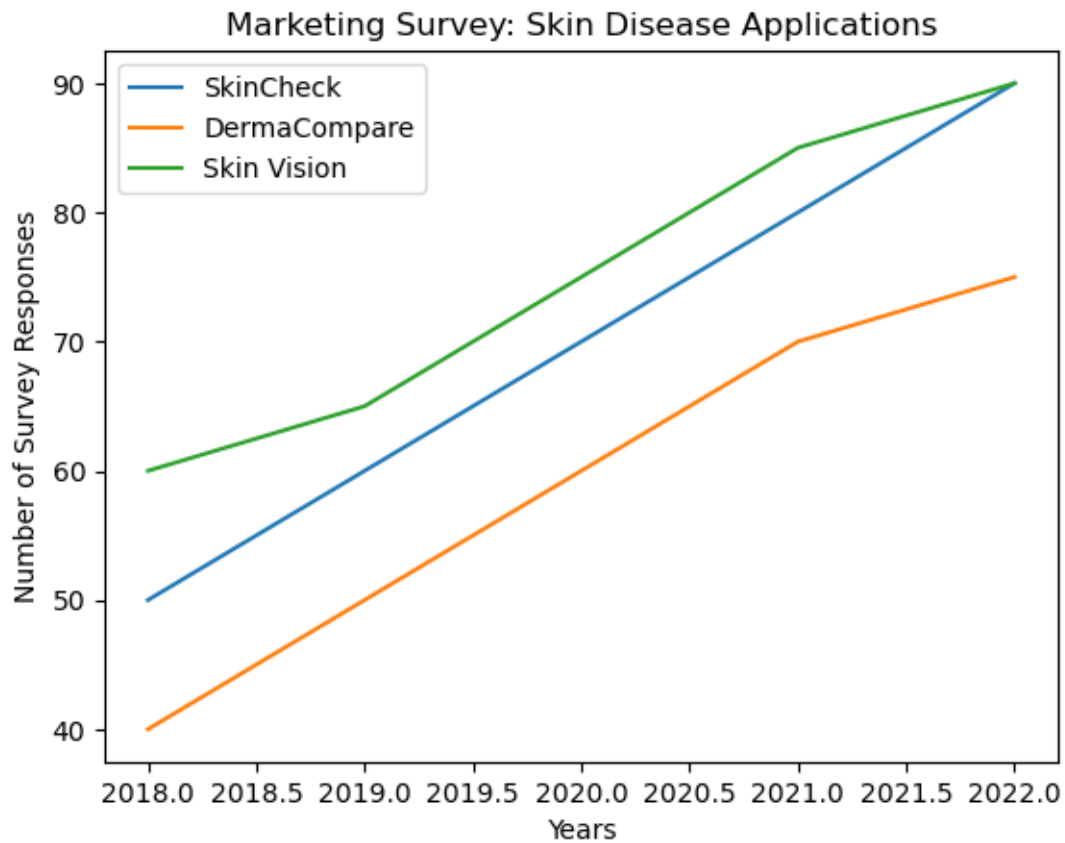


Figure 1 Marketing Survey

2.7 Research Gap

Although recent research has made significant advancements in using deep learning techniques for skin disease classification, research gaps still need to be addressed. One of these gaps pertains to the limited availability of diverse and representative datasets for training and evaluating skin disease classification models. Another gap involves exploring effective data augmentation techniques specifically tailored for skin disease classification.

Existing studies in skin disease classification often rely on publicly available datasets such as ISIC-2019, HAM10000, and PH2. While these datasets provide valuable resources for research. These datasets predominantly consist of images from lighter-skinned individuals,

which may limit the generalization of models to individuals with darker skin tones. Additionally, there is a need for more comprehensive datasets.

Research efforts should focus on collecting larger and more diverse datasets encompassing a broader range of skin diseases and include images from individuals with different skin tones and ethnic backgrounds. By doing so, the resulting models can be more robust and reliable in real-world scenarios, where skin diseases may manifest differently across populations.

Data augmentation plays a crucial role in mitigating the limitations posed by limited datasets. By applying various augmentation techniques, researchers can artificially increase the size of the dataset, introduce diversity, and improve the generalization capabilities of skin disease classification models. However, the existing research often lacks a comprehensive exploration of augmentation techniques specifically tailored for skin disease classification.

Addressing these research gaps related to limited datasets and augmentation techniques in skin disease classification can significantly enhance the accuracy, generalization, and reliability of deep learning models for diagnosing and classifying skin disorders. Furthermore, these advancements would pave the way for more inclusive and effective dermatological solutions that cater to diverse populations and address the challenges associated with skin disease diagnosis.

2.8 Problem Statement

Skin diseases present significant challenges regarding accurate diagnosis and timely treatment, impacting millions of individuals worldwide. The visual similarity among various skin conditions and patients' hesitation to seek medical advice often lead to diagnostic uncertainties and delayed interventions. This lack of accessible and reliable diagnostic tools hampers the ability to provide timely and appropriate care, potentially resulting in prolonged suffering, worsening symptoms, and decreased quality of life for patients. Moreover, geographical and logistical barriers restrict access to specialized dermatological care, particularly for individuals residing in remote areas or with limited mobility. The current healthcare system lacks efficient mechanisms to support dermatologists in making accurate diagnoses and facilitating seamless patient-doctor interactions. So an innovative solution is urgently needed that harnesses advanced technologies, such as deep learning algorithms and user-friendly interfaces, to address these challenges and provide reliable, accessible, and timely diagnoses of skin diseases. By doing so, this solution would not only improve treatment outcomes but also enhance patient satisfaction and overall well-being.

In summary, the key problems faced in the field of skin disease diagnosis are:

- **Diagnostic Uncertainty:** The visual similarities among different skin conditions frequently lead to difficulty accurately diagnosing specific diseases. This results in diagnostic uncertainties, prolonged diagnostic processes, and delays in initiating appropriate treatment.
- **Limited Access to Dermatological Care:** Geographical and logistical barriers hinder access to specialized dermatological care, particularly for individuals in remote areas or with limited mobility. This lack of accessibility results in delayed diagnoses, limited treatment options, and reduced opportunities for early intervention.
- **Inadequate Diagnostic Support Tools:** The current healthcare system lacks efficient tools and resources that can assist dermatologists in accurately diagnosing skin diseases. This leads to variations in diagnostic accuracy and potentially compromises the quality of patient care.
- **Insufficient Patient-Doctor Interactions:** Inefficient communication and limited interaction between patients and dermatologists impede effective care delivery. Challenges in scheduling appointments, accessing timely consultations, and sharing relevant information and resources hinder collaborative decision-making and can negatively impact patient outcomes.
- **Inadequate Training and Expertise:** Not all healthcare professionals receive extensive training in dermatology, leading to variations in diagnostic accuracy and potentially missed or misdiagnosed skin conditions. A lack of specialized expertise can result in ineffective treatment plans and compromised patient care.

Addressing these problems necessitates developing and implementing an innovative solution that leverages advanced technologies, such as deep learning algorithms.

Chapter 3: Requirements and Design

3.1 Requirements

This section outlines the requirements for the development of the SkinCheck AI system. The requirements are categorized into functional and non-functional requirements, ensuring a comprehensive understanding of the system's capabilities and performance expectations.

3.1.1 Functional Requirements

The functional requirements of the SkinCheck AI system outline the specific features and functionalities essential for the system to effectively fulfill its purpose and meet the needs of its users. These requirements describe the desired behaviors and interactions of the system, ensuring that it provides a seamless and user-friendly experience. The following are the functional requirements of the SkinCheck AI:

- FR.1:** Patients or doctors shall be able to signup and login to their accounts.
- FR.2:** Patients or doctors can edit and manage their profile information.
- FR.3:** Patients shall be able to search for dermatologists.
- FR.4:** Patients can view dermatologists' profiles, including qualifications, experience, and patient reviews.
- FR.5:** The patient shall be able to book appointments for video consultations.
- FR.6:** Patients can securely share images of their skin condition with dermatologists during online consultations.
- FR.7:** The patient shall be able to pay securely before booking appointments, ensuring a convenient and hassle-free payment option for transactions.
- FR.8:** Patients and doctors shall receive timely reminders about upcoming appointments through push notifications.
- FR.9:** The app lets Patients view and download dermatology prescriptions and medical records.
- FR.10:** Patients shall be able to provide feedback and ratings for dermatologists they have consulted.
- FR.11:** Doctors shall have a calendar view to manage appointments, including accepting or rejecting appointment requests, rescheduling appointments, and marking appointments as completed.
- FR.12:** Doctors shall be able to view and manage patient records, including medical history,

prescribed medications, and follow-up appointments.

FR.13: Doctors can conduct online consultations with patients through video calls.

FR.14: Doctors shall be able to generate and send electronic prescriptions to patients, including dosage instructions and medicine recommendations.

FR.15: Doctors shall be able to detect skin disease by uploading images of the patient's skin condition.

FR.16: Doctors shall be able to request withdrawals of their earnings securely and conveniently.

FR.17: Doctors shall have access to analytics and reports, including appointment statistics, patient feedback, and ratings.

FR.18: Doctors can share educational resources, such as articles, videos, or other informative content with their patients.

FR.19: The administrator shall have the ability to manage user accounts.

FR.20: The administrator shall verify the credentials and qualifications of doctors during the registration process.

FR.21: The administrator shall have access to the appointment system to manage and resolve conflicts or issues.

3.1.2 Non-Functional Requirements

N-FRs represent the criteria that judge the operation of a system rather than specific behaviors. They are designed to ensure the system is user-friendly, reliable, and efficient.

3.1.3 Hardware and Software Requirements

The successful implementation and functioning of the SkinCheck AI system depend on specific hardware and software requirements. These requirements ensure the system operates efficiently and smoothly, providing users with a seamless experience while accessing the application's features and services. The following are hardware and software requirements:

3.1.3.1 Hardware Requirements

HR.1: Minimum 2GB RAM for mobiles and minimum 4GB RAM for laptops.

HR.2: A quad-core processor or higher for mobiles and a dual-core or higher for laptops.

HR.3: High-resolution display (720p or higher) for mobiles and display resolution of 1080p or higher for laptops.

HR.4: Reliable internet connectivity.

HR.5: Sufficient storage space.

3.1.3.2 Software Requirements

HR.6: Android or iOS Operating System for mobile and Windows, macOS, or Linux for laptop.

HR.7: Minimum 3G Internet connection (recommended 4G or higher).

HR.8: (Browser support) Latest version of Chrome, Firefox, Safari, or Edge.

3.2 Proposed Methodology

The proposed methodology for developing the SkinCheck AI system involves steps to ensure better skin disease detection, a convenient user experience, and reliable performance. The methodology encompasses data collection, preprocessing, model selection, training and evaluation, integration and deployment, testing and validation.

- **Data Collection:** Collect a diverse and extensive dataset of skin disease images with corresponding labels. Ensure the inclusion of various skin types, ages, and disease types to train the algorithm effectively. For example, the HAM10000 dataset for melanoma can be utilized.
- **Data Preprocessing:** Preprocess the collected data to convert it into usable information. Ensure the quality of images and labels, addressing any biases or inconsistencies. Apply data augmentation techniques, such as flipping, rotating, and zooming, to increase the diversity of the dataset and prevent overfitting.
- **Model Selection:** Choose an appropriate convolutional neural network (CNN) architecture for image classification, such as EfficientNet, which has shown promising results in various computer vision tasks. Consider factors like model complexity, computational resources, and the ability to handle skin disease detection accurately.
- **Model Training and Evaluation:** Train the selected CNN model using the preprocessed dataset. Separate the dataset into training, validation, and testing sets for model evaluation. Optimize the model's hyperparameters through techniques like learning rate tuning and regularization. Evaluate the model's performance on the testing set, measuring accuracy, precision, recall, and F1 score metrics.
- **Integration:** Develop the front end of the SkinCheck AI website using React.js, ensuring an intuitive user interface for dermatologists and patients. Implement the backend using Node.js to handle user authentication, image analysis, diagnosis, and secure communication. Integrate the trained CNN model into the backend, enabling automated skin disease detection and diagnosis based on uploaded images. Implement secure data transmission and storage protocols to protect user information and medical data.

- **Testing and Validation:** Conduct thorough testing and validation of the SkinCheck AI system to ensure its accuracy and reliability. To assess its effectiveness, compare the system's diagnoses against expert dermatologists' opinions and known ground truth labels. Gather feedback from users, dermatologists, and patients, to refine and improve the system based on real-world usage scenarios.

3.3 System Architecture

The system architecture of the SkinCheck AI website is designed to facilitate seamless communication between users (patients and dermatologists) and ensure effective skin disease detection. It comprises multiple components working together to provide a reliable and user-friendly experience. The architecture includes the front and back end, machine learning components, data storage, integration, and deployment layers.

▪ **Frontend:**

- The frontend is developed using React.js, a popular JavaScript library for building user interfaces.
- It includes user interface components that allow users to interact with the website, upload images, and receive diagnosis results.
- The frontend communicates with the backend to send image data for analysis.

▪ **Backend:**

- The backend is built using Node.js, providing a runtime environment for executing server-side JavaScript code.
- It handles user authentication, image processing, diagnosis, and secure communication between the frontend and machine learning components.
- The backend interacts with the machine learning component to process the uploaded images and obtain skin disease predictions.

▪ **Machine Learning Component:**

- The machine learning component incorporates a convolutional neural network (CNN) model specifically trained for skin disease detection.
- The CNN model analyzes the uploaded images and generates predictions regarding the presence of skin diseases.
- The model utilizes image classification techniques to identify patterns and features indicative of various skin conditions.

▪ **Data Storage:**

Requirements and Design

- The system employs a database management system, such as SQL Server or MongoDB, to securely store user data, image metadata, and diagnosis results.
- Images uploaded by users are stored in a secure and scalable storage solution, ensuring data privacy and accessibility.
- **Integration:**
 - Integration with external services, such as video consultation platforms or notification systems, can be implemented using APIs to enhance the user experience and communication between dermatologists and patients.

3.4 Use Cases

3.4.1 Patient Registration and Login

Table 2: Patient Registration and Login

Name		Patient Registration and Login	
Actors		Patient	
Summary		The patient can register and log in to the system.	
Pre-Conditions		The patient must have a valid email address.	
Post-Conditions		The patient's account is created and they are logged in.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The patient provides their name, contact details, and email address.	2	The system creates a new account and confirms the registration.
3	The patient logs in using their preferred login information.	4	The system authenticates the patient and logs them in.
Alternative Flow			
3	The patient forgets their login information.	4-A	The system provides a "Forgot Password" option and guides the patient through the process of resetting their password.

3.4.2 Profile Management

Table 3: Profile Management

Name		Profile Management	
Actors		Patient	

Summary		The patient can edit and manage their profile information.	
Pre-Conditions		The patient must be logged in.	
Post-Conditions		The patient's profile information is updated.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The patient accesses their profile.	2	The system displays the patient's profile.
3	The patient updates their contact details and other information.	4	The system saves the updated information.
Alternative Flow			
3	The patient wants to delete their account.	4-A	The system provides an option to delete the account and confirms the action with the patient before proceeding.

3.4.3 Search and Book Appointments

Table 4: Search and Book Appointments

Table 4: Search and Book Appointments			
Name		Search and Book Appointments	
Actors		Patient	
Summary		The patient can search for dermatologists and book appointments.	
Pre-Conditions		The patient must be logged in.	
Post-Conditions		The appointment is booked with the selected dermatologist.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The patient searches for dermatologists based on various filters.	2	The system displays a list of dermatologists matching the criteria.
3	The patient views dermatologists' profiles and selects one.	4	The system displays the selected dermatologist's profile.
4	The patient books an appointment with the selected dermatologist.	5	The system confirms the appointment booking.
Alternative Flow			
4	The patient wants to cancel a booked appointment.	5-A	The system provides an option to cancel the appointment and confirms the action with the patient before proceeding.

3.4.4 Doctor Profile Management

Table 5: Doctor Profile Management

Name		Doctor Profile Management	
Actors		Doctor	
Summary		The doctor can view and edit their profile information	
Pre-Conditions		The doctor must be logged in.	
Post-Conditions		The doctor's profile information is updated in the system.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The doctor navigates to their profile page.	2	The profile page is displayed, showing the doctor's information.
3	The doctor updates their profile information and clicks the save button.	4	The system saves the updated profile information and displays a success message.
Alternative Flow			
3	The doctor cancels the changes.	4-A	The system discards the changes and displays the original profile information.

3.4.5 Doctor Registration and Profile Creation

Table 6: Doctor Registration and Profile Creation

Table 17: Doctor Registration and Profile Creation			
Name		Doctor Registration and Profile Creation	
Actors		Doctor	
Summary		The doctor can register and create their profile.	
Pre-Conditions		The doctor must have valid credentials and qualifications.	
Post-Conditions		The doctor's profile is created and verified.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The doctor provides their qualifications, specialization, clinic/hospital details, available time slots, and consultation fees.	2	The system creates a new profile and initiates the verification process.
3	The doctor logs in to their account.	4	The system authenticates the doctor and logs them in.

Alternative Flow			
3	The doctor forgets their login information.	4-A	The system provides a "Forgot Password" option and guides the doctor through the process of resetting their password.

3.4.6 Online Consultations

Table 7: Online Consultations

Table 7: Online Consultations			
Name		Online Consultations	
Actors		Doctor, Patient	
Summary		The doctor and patient can conduct online consultations through video calls or chat.	
Pre-Conditions		The doctor and patient must be logged in. An appointment for an online consultation must be scheduled.	
Post-Conditions		The online consultation is completed.	
Special Requirements		The system must have a secure and reliable video call and chat feature.	
Basic Flow			
Actor Action		System Response	
1	The doctor and patient navigate to the online consultation section at the scheduled time.	2	The system initiates the online consultation through a video call or chat.
3	The doctor and patient conduct the consultation, discussing the patient's condition and treatment options.	4	The system facilitates the video call or chat, allowing the doctor and patient to share documents and images as needed.
Alternative Flow			
3	The doctor or patient experiences technical issues during the online consultation.	4-A	The system provides troubleshooting options and allows the doctor or patient to reconnect to the consultation.

3.4.7 Appointment Management

Table 8: Appointment Management

Name		Appointment Management	
Actors		Doctor	
Summary		The doctor can manage their appointments, including accepting or rejecting appointment requests, rescheduling appointments, and marking appointments as completed.	
Pre-Conditions		The doctor must be logged in. There must be appointments scheduled.	
Post-Conditions		The appointments are managed as per the doctor's actions.	
Special		None	

Requirements			
Basic Flow			
Actor Action		System Response	
1	The doctor navigates to the appointment management section.	2	The system displays a calendar view of the doctor's appointments.
3	The doctor accepts or rejects appointment requests, reschedules appointments, or marks appointments as completed.	4	The system updates the appointments based on the doctor's actions.
Alternative Flow			
3	The doctor wants to send a notification to a patient about a change in appointment.	4-A	The system provides an option to send a notification and sends it upon confirmation from the doctor.

3.4.8 Skin Disease Detection

Table 9: Skin Disease Detection

Table 3: Skin Disease Detection			
Name		Skin Disease Detection	
Actors		Doctor, Patient	
Summary		The doctor can check skin disease detection by uploading images of the patient's skin condition.	
Pre-Conditions		The doctor must be logged in. The patient must have shared images of their skin condition.	
Post-Conditions		The doctor receives a disease name.	
Special Requirements		The system must have a built-in or integrated AI model capable of analyzing skin conditions based on images.	
Basic Flow			
Actor Action		System Response	
1	The doctor uploads an image of the patient's skin condition.	2	The system analyzes the image using its AI model.
3	The doctor can view the disease name.	4	The system will display disease name and give option to check another image.
Alternative Flow			
3	The system cannot analyze the image due to poor quality or other issues.	4-A	The system notifies the doctor about the issue and requests a new image.

3.4.9 Payment Processing

Table 10: Payment Processing

Name	Payment Processing
Actors	Patient

Summary		The patient can pay for their appointments.	
Pre-Conditions		The patient must be logged in and have a payment method linked to their account.	
Post-Conditions		The payment is successfully processed, and the patient receives a receipt.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The patient navigates to the payment page.	2	The payment page is displayed, allowing the patient to select a payment method.
3	The patient selects their payment method and enters payment details.	4	The system processes the payment and generates a receipt for the patient.
Alternative Flow			
3	The patient enters invalid payment details.	4-A	The system responds with an error message: "Invalid payment details, please try again."

3.4.10 Withdrawal of Earnings

Table 11: Withdrawal of Earnings

Table 11: Withdrawal of Earnings			
Name		Withdrawal of Earnings	
Actors		Doctor	
Summary		The doctor can request withdrawals of their earnings securely and conveniently.	
Pre-Conditions		The doctor must be logged in. The doctor must have earnings in their account.	
Post-Conditions		The withdrawal request is processed.	
Special Requirements		The system must have a secure and reliable payment gateway for processing withdrawals.	
Basic Flow			
Actor Action		System Response	
1	The doctor navigates to the earnings section.	2	The system displays the doctor's total earnings and the option to request a withdrawal.
3	The doctor requests a withdrawal of a certain amount.	4	The system processes the withdrawal request and updates the doctor's earnings balance.
Alternative Flow			
3	The doctor requests a withdrawal that exceeds their earnings balance.	4-A	The system notifies the doctor about the issue and does not process the withdrawal request.

3.4.11 Rate and Review

Table 12: Rate and Review

Name	Rate and Review		
Actors	Patient		
Summary	The patient can rate and provide feedback for dermatologists they have had appointments with.		
Pre-Conditions	The patient must have had an appointment with the dermatologist.		
Post-Conditions	The dermatologist's rating and feedback are updated.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The patient navigates to the dermatologist's profile page.	2	The system displays the dermatologist's profile page.
3	The patient selects the "Rate and Review" button.	4	The system displays a form for the patient to rate and provide feedback.
Alternative Flow			
3	The patient provides their rating and feedback and submits the form.	4-A	The system updates the dermatologist's rating and displays the feedback on the profile page.

3.4.12 User Management

Table 13: User Management

Table 13: User Management			
Name		User Management	
Actors		Admin	
Summary		The admin can manage user accounts.	
Pre-Conditions		The admin must be logged in.	
Post-Conditions		The user accounts are managed as per the admin's actions.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The admin accesses the user management section.	2	The system displays a list of user accounts.
3	The admin performs management actions on user accounts, such as approving registration or suspending accounts.	4	The system updates the user accounts based on the admin's actions.

Alternative Flow			
3	The admin sends a notification to all users.	4-A	The system provides an option to send a mass notification and sends it upon confirmation from the admin.

3.4.13 Doctor Verification

Table 14: Doctor Verification

Table 14: Doctor Verification			
Name		Doctor Verification	
Actors		Admin	
Summary		The admin verifies the credentials and qualifications of doctors.	
Pre-Conditions		The doctor must have registered and provided their credentials and qualifications.	
Post-Conditions		The doctor's profile is verified.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The admin accesses the doctor verification section.	2	The system displays a list of doctors awaiting verification.
3	The admin verifies the credentials and qualifications of the doctors.	4	The system updates the doctors' profiles as verified.
Alternative Flow			
3	The admin finds discrepancies in a doctor's credentials or qualifications.	4-A	The system provides an option to reject the doctor's registration and sends a notification to the doctor about the rejection.

3.4.14 Appointment Management

Table 15: Appointment Management

Name		Appointment Management	
Actors		Admin	
Summary		The admin can manage and resolve any conflicts or issues that may arise in the appointment system.	
Pre-Conditions		There must be appointments scheduled in the system.	
Post-Conditions		Any conflicts or issues in the appointment system are resolved.	
Special Requirements		None	

Requirements and Design

Basic Flow			
Actor Action		System Response	
1	The admin accesses the appointment management section.	2	The system displays a list of scheduled appointments.
3	The admin manages and resolves any conflicts or issues in the appointment system.	4	The system updates the appointments based on the admin's actions.
Alternative Flow			
3	The admin sends a notification to a patient or doctor about a change in appointment.	4-A	The system provides an option to send a notification upon confirmation from the admin.

3.5 Methodology Diagram

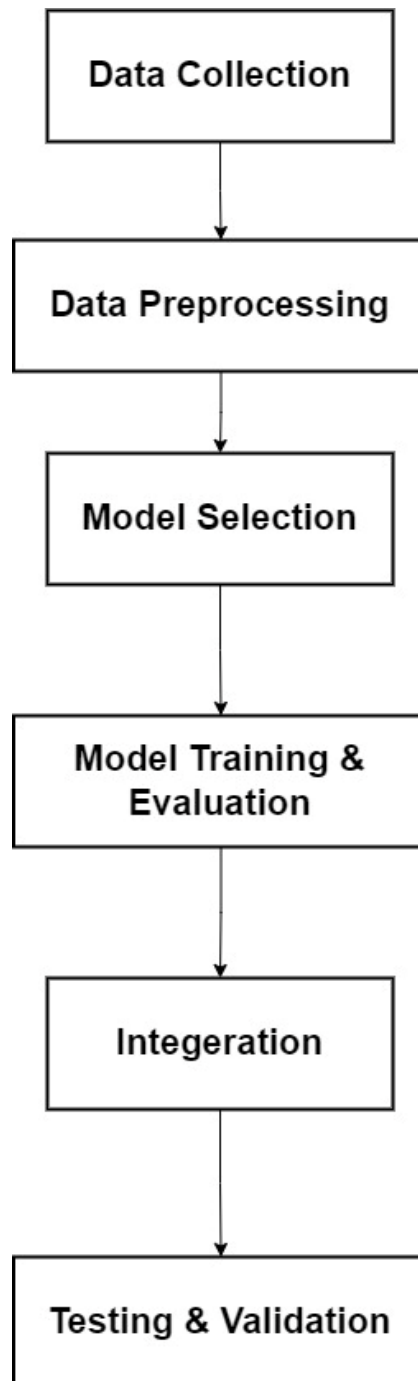


Figure 2: Methodology Diagram

3.6 Sequence diagram

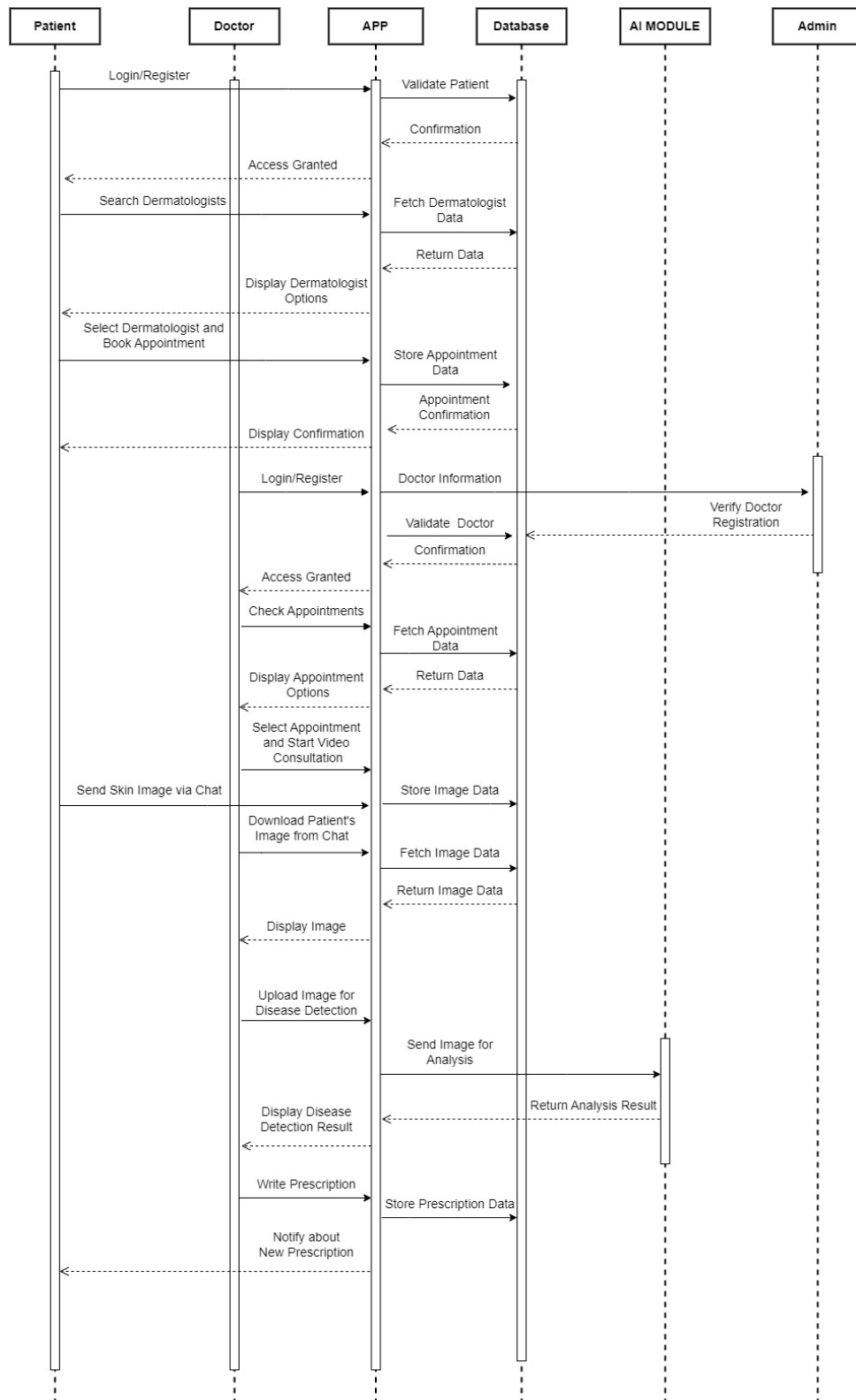


Figure 3: Sequence Diagram

3.7 Use Case Diagram

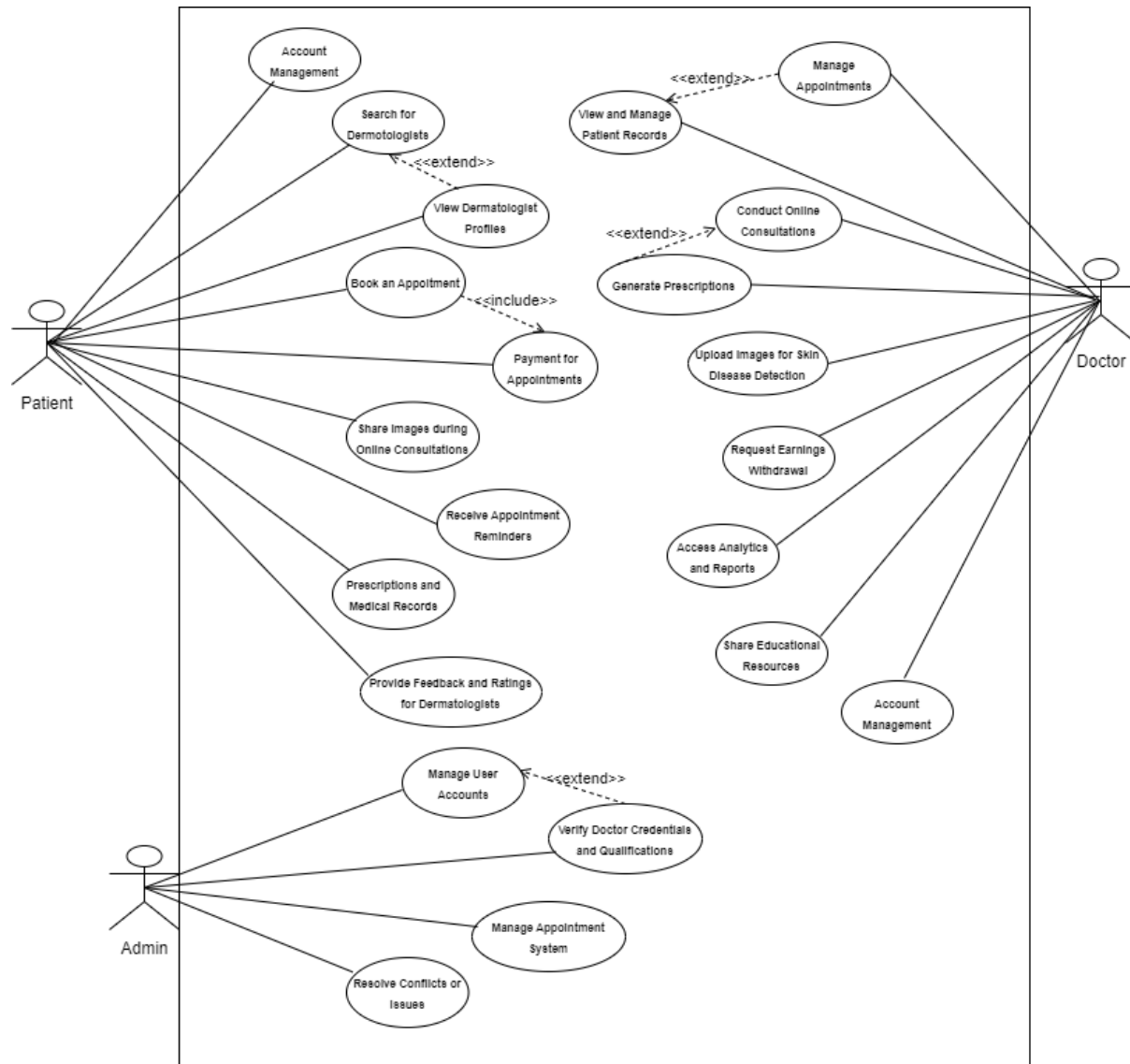


Figure 4 Use Case Diagram

3.8 CNN Architecture Diagram

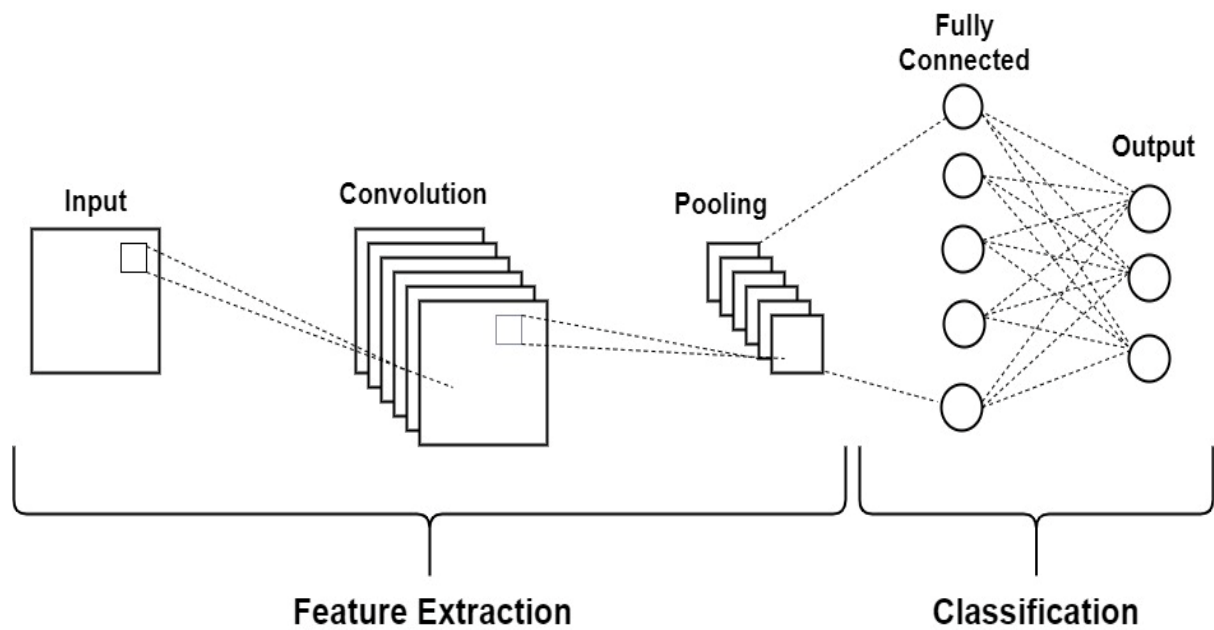
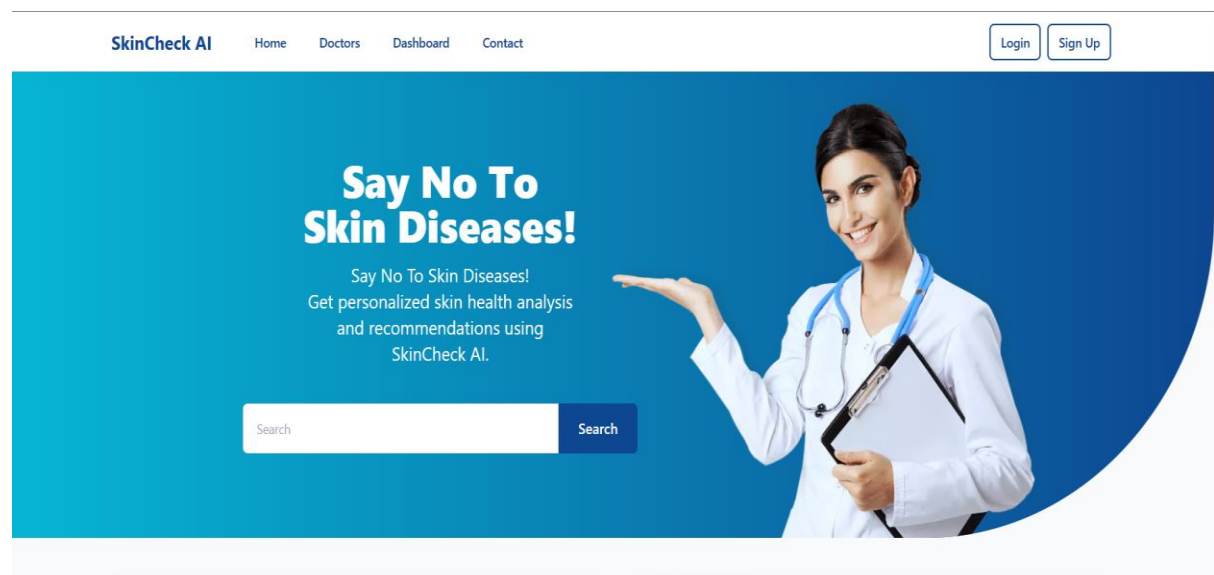
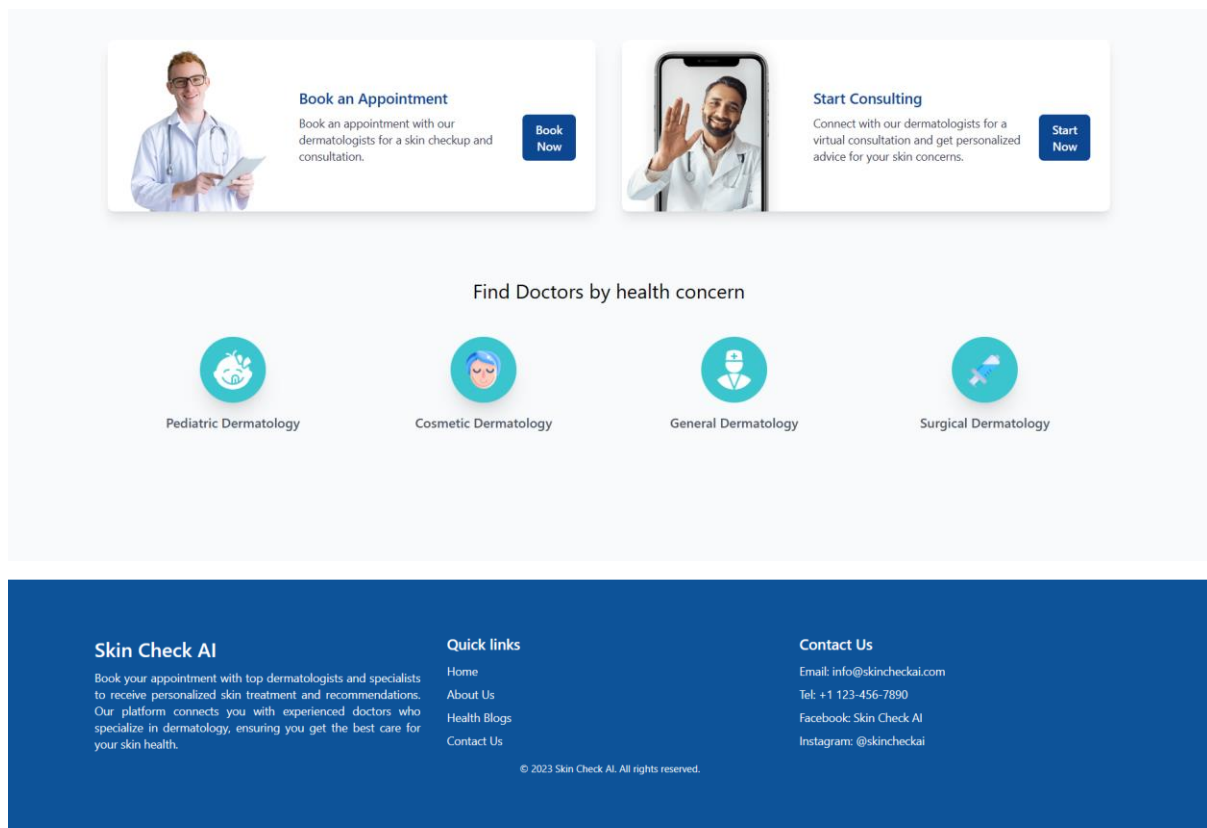


Figure 5: CNN Architecture Diagram

3.9 GUI Graphical User Interfaces

3.9.1 Home Page





The homepage of our skin disease detection and video consultation booking website serves as a comprehensive platform catering to both patients and doctors. Upon accessing the homepage, users are greeted with an intuitive and user-friendly interface designed to provide a seamless experience. The primary focus of the homepage is to facilitate easy navigation and access to various modules, which can only be accessed by registered users. A prominent search feature allows users to find doctors specializing in various categories such as Cosmetic Dermatology, General Dermatology, and Surgical Dermatology. This empowers patients to locate the right experts for their specific needs. Furthermore, the video consultation module enables convenient virtual appointments, allowing patients to connect with dermatologists remotely. The homepage also includes sections dedicated to contacting us and accessing the login and registration portals for both patients and doctors. Overall, the homepage provides a centralized hub for individuals seeking effective skin disease detection and reliable video consultation services.

3.9.2 Login

3.9.2.1 Patient Login

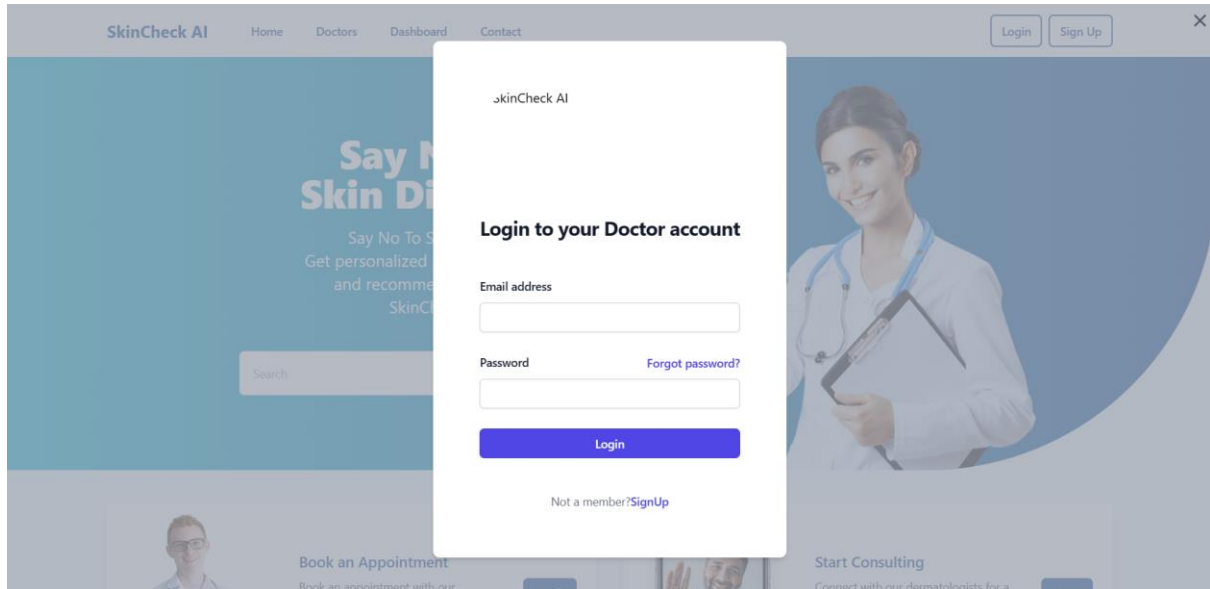
The screenshot displays the SkinCheck AI website's patient login interface. A central white modal is overlaid on a blurred background of the website. The modal is titled "Login to your Patient account" and contains the following elements:

- Header: "SkinCheck AI"
- Form fields: "Email address" and "Password" (with a "Forgot password?" link).
- Buttons: "Login" (blue) and "Sign Up" (purple).
- Footer: "Not a member? Sign Up"

The background website header includes navigation links: "Home", "Doctors", "Dashboard", and "Contact". The hero section features a large image of a female doctor and text: "Say No To Skin Disease", "Get personalized recommendations and recommendations from SkinCheck AI", and a "Search" button.

The login page of our skin disease detection and video consultation booking website offers a secure and streamlined process for patients to access their accounts. The page presents a clean and intuitive layout, prioritizing user convenience and privacy. At the center of the login page, there are prominently displayed fields where patients can enter their registered email address and password. The email field ensures accurate identification, while the password field safeguards account confidentiality. Furthermore, there is a Signup option for the patients if they want to Signup.

3.9.2.2 Doctor Login



The screenshot displays the SkinCheck AI website with a modal window for doctor login. The modal is titled "Login to your Doctor account" and contains the following elements:

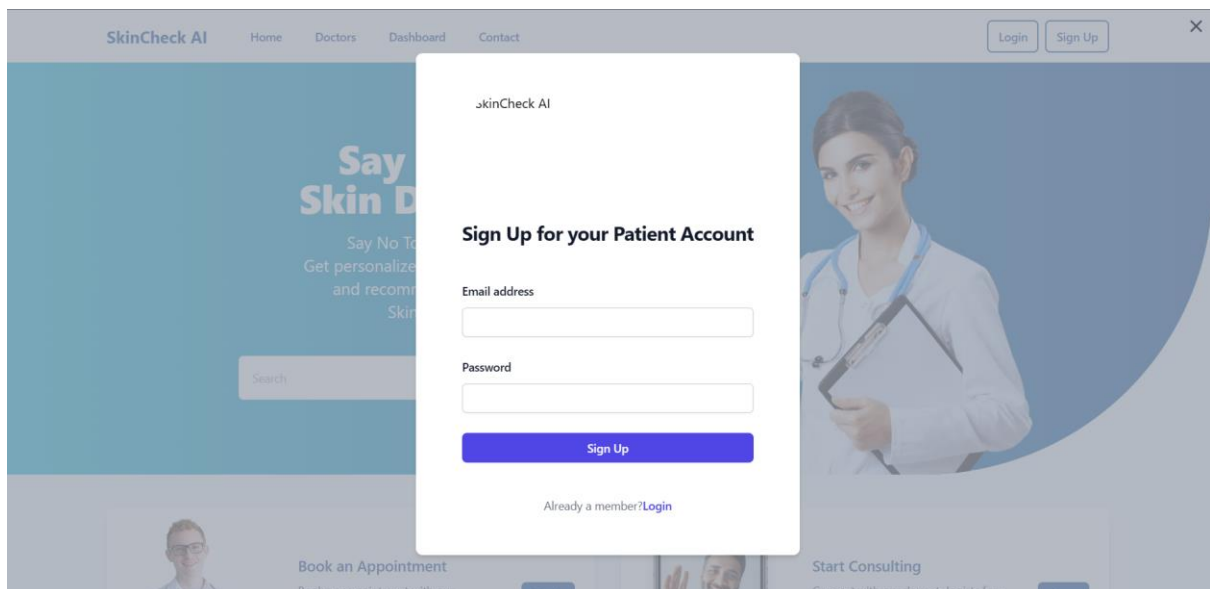
- Header: SkinCheck AI
- Form fields: Email address and Password.
- Links: [Forgot password?](#)
- Buttons: [Login](#) (blue) and [Sign Up](#) (purple).
- Footer: Not a member? [Sign Up](#)

The background of the website shows a navigation bar with "Home", "Doctors", "Dashboard", and "Contact" links, and a hero section with the text "Say No To Skin Disease" and "Get personalized and recommended SkinCheck AI".

The login page of our skin disease detection and video consultation booking website offers a secure and streamlined process for Doctors to access their accounts. The page presents a clean and intuitive layout, prioritizing user convenience and privacy. At the center of the login page, there are prominently displayed fields where Doctors can enter their registered email address and password. The email field ensures accurate identification, while the password field safeguards account confidentiality. Furthermore, there is a [Sign Up](#) option for the Doctors if they want to [Sign Up](#).

3.9.3 Sign Up

3.9.3.1 Patient Sign Up

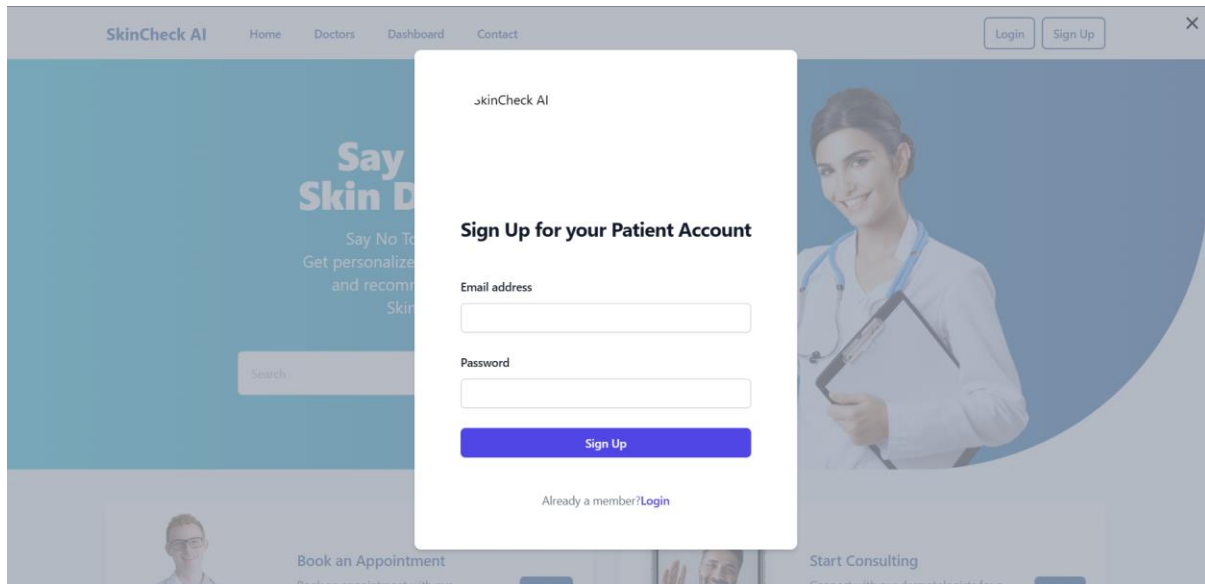


The screenshot displays the SkinCheck AI website with a modal window for patient sign up. The modal is titled "Sign Up for your Patient Account" and contains the following elements:

- Header: SkinCheck AI
- Form fields: Email address and Password.
- Buttons: [Sign Up](#) (blue) and [Login](#) (purple).
- Footer: Already a member? [Login](#)

The background of the website shows a navigation bar with "Home", "Doctors", "Dashboard", and "Contact" links, and a hero section with the text "Say No To Skin Disease" and "Get personalized and recommended SkinCheck AI".

3.9.3.2 Doctor SignUp



The screenshot displays the SkinCheck AI website with a modal form for patient sign-up. The modal is titled "Sign Up for your Patient Account" and includes the following elements:

- Header:** "SkinCheck AI" logo.
- Form Fields:**
 - "Email address" with a text input field.
 - "Password" with a text input field.
- Buttons:** A blue "Sign Up" button.
- Footer:** A link that says "Already a member? Login".

The background of the website shows a navigation bar with links: Home, Doctors, Dashboard, and Contact. There are also "Login" and "Sign Up" buttons in the top right corner. The main content area features a large image of a female doctor and text promoting skin disease detection and video consultations.

Both of the pages Doctor SignUp and Patient Signup provide a seamless interface for them to SignUp and Register their accounts by providing emails and password.

3.9.4 Patient Dashboard

The Patient Dashboard of our skin disease detection and video consultation booking website offers patients a comprehensive and user-friendly platform to manage their account and access various essential features. The dashboard serves as a central hub where patients can navigate through different options in the navigation bar, enabling them to effectively utilize the website's functionalities.

3.9.4.1 Profile

The screenshot displays the 'Profile' section of the SkinCheck AI Patient Dashboard. The top navigation bar includes 'SkinCheck AI', 'Home', 'Doctors', 'Dashboard', and 'Contact', along with 'Login' and 'Sign Up' buttons. The left sidebar, titled 'Patient Dashboard', lists 'Profile' (selected), 'Appointments', 'Settings', 'Medical Records', 'Messages', 'Billing', and 'Logout'. The main content area, titled 'Profile', features a circular profile picture of a doctor, an 'Upload photo' button, and a note about accepted image formats (PNG, GIF, JPEG). Below this are input fields for 'Full Name' (Abiad Daud), 'Email' (abc@gmail.com), 'Address' (123 Main St, City), and 'Date of Birth' (01/01/1980). A 'Save Changes' button is positioned at the bottom of the form. The footer contains three sections: 'Skin Check AI' with a description of the service, 'Quick links' (Home, About Us, Health Blogs, Contact Us), and 'Contact Us' with email, phone, and social media information. A copyright notice for 2023 is also present.

Profile

Upload photo
Accepted formats: PNG, GIF, JPEG

Full Name
Abiad Daud

Email
abc@gmail.com

Address
123 Main St, City

Date of Birth
01/01/1980

Save Changes

Skin Check AI
Book your appointment with top dermatologists and specialists to receive personalized skin treatment and recommendations. Our platform connects you with experienced doctors who specialize in dermatology, ensuring you get the best care for your skin health.

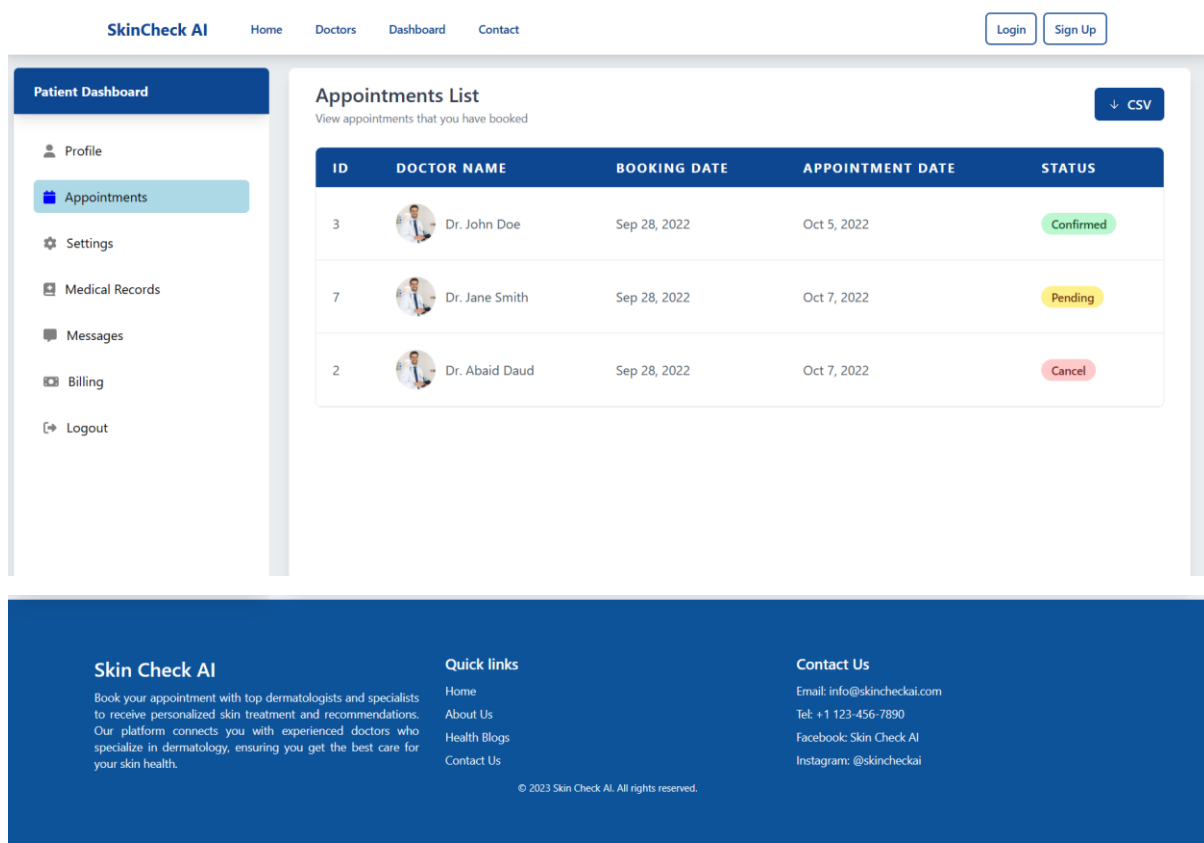
Quick links
Home
About Us
Health Blogs
Contact Us

Contact Us
Email: info@skincheckai.com
Tel: +1 123-456-7890
Facebook: Skin Check AI
Instagram: @skincheckai

© 2023 Skin Check AI. All rights reserved.

The "Profile" option in the navigation bar allows patients to update their profile information easily. By selecting this option, patients can modify details such as their profile picture, name, date of birth (DOB), and email address. This ensures that patients can maintain accurate and up-to-date personal information within their accounts, enhancing communication and personalization.

3.9.4.2 Appointments



Another key option in the navigation bar is "Appointments." When patients click on this option, they can view a comprehensive list of their past and future appointments. The displayed information includes the name of the doctor, the date and time of the appointment, and the status (e.g., confirmed, pending, completed). To facilitate record keeping and analysis, patients can download their appointment data in CSV format by clicking on the "CSV" option.

3.9.4.3 Settings

SkinCheck AI

HomeDoctorsDashboardContact

LoginSign Up

Patient Dashboard

Profile

Appointments

Settings

Medical Records

Messages

Billing

Logout

Account settings

Email Address

Your email address is john.doe@company.com

Change

Password

Current Password

New Password

Save Password

Skin Check AI

Book your appointment with top dermatologists and specialists to receive personalized skin treatment and recommendations. Our platform connects you with experienced doctors who specialize in dermatology, ensuring you get the best care for your skin health.

Quick links

Home

About Us

Health Blogs

Contact Us

Contact Us

Email: info@skincheckai.com

Tel: +1 123-456-7890

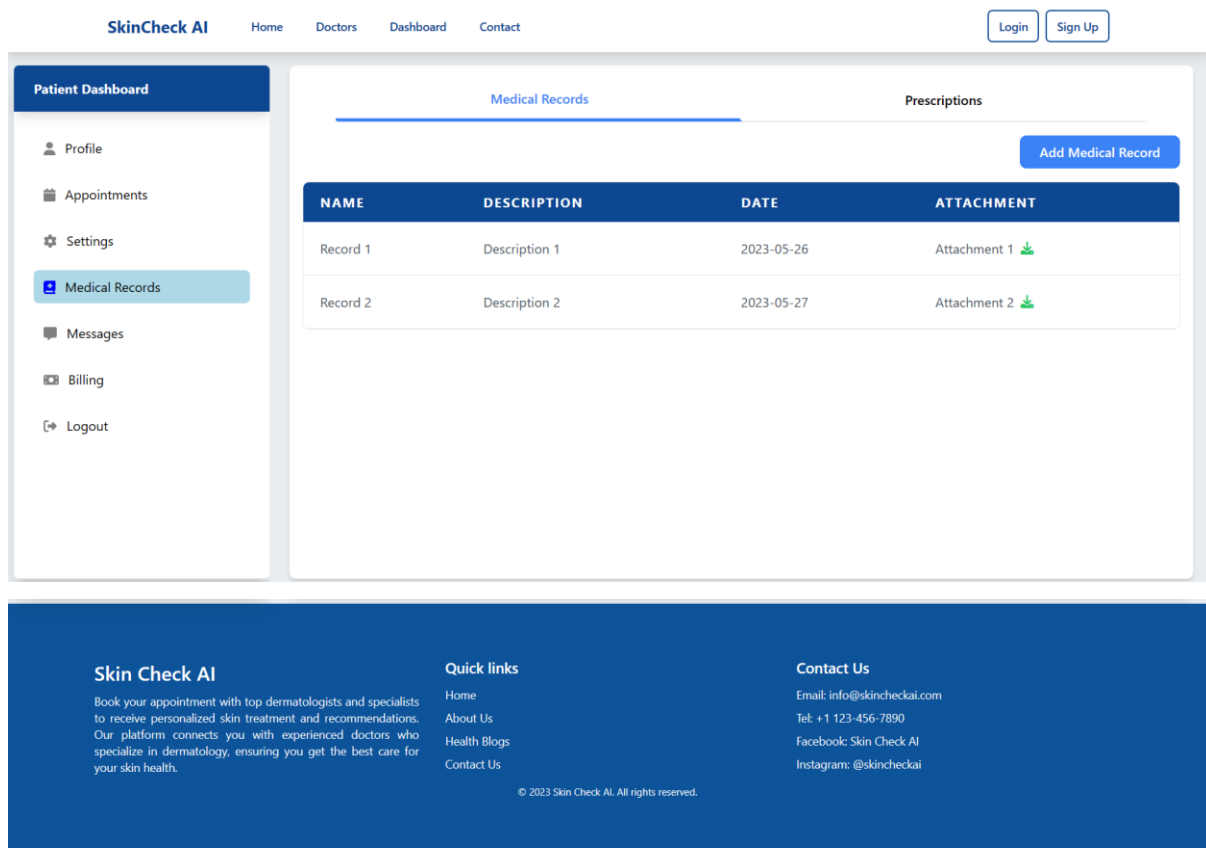
Facebook: Skin Check AI

Instagram: @skincheckai

© 2023 Skin Check AI. All rights reserved.

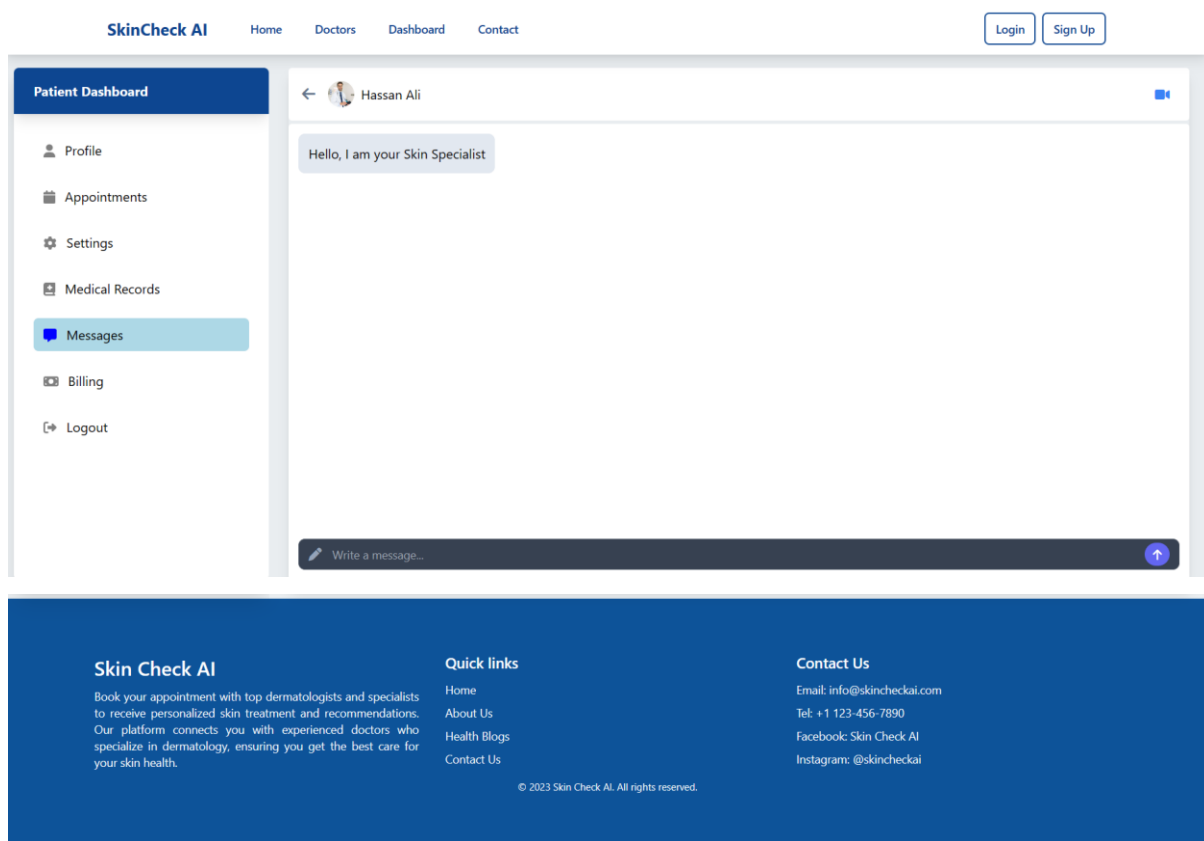
The "Settings" option empowers patients to have control over their account's security. Patients can click on this option to access a secure page where they can change their account password. This feature ensures that patients can easily update their password as needed, enhancing account protection.

3.9.4.4 Medical Records



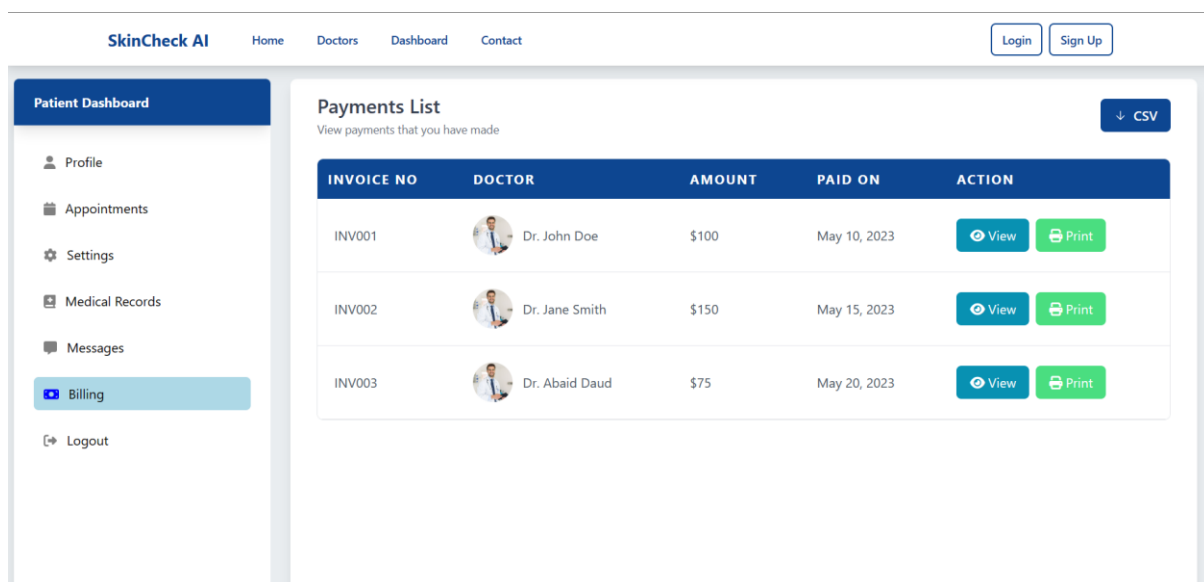
The "Medical Record" option in the navigation bar allows patients to add and manage their medical records. Patients can add new medical records, which will facilitate doctors in providing personalized and effective treatments. Furthermore, patients can switch to the "Prescription" option, where they can view prescriptions provided by doctors, making it convenient to access and refer to prescribed medications and instructions.

3.9.4.5 Messages

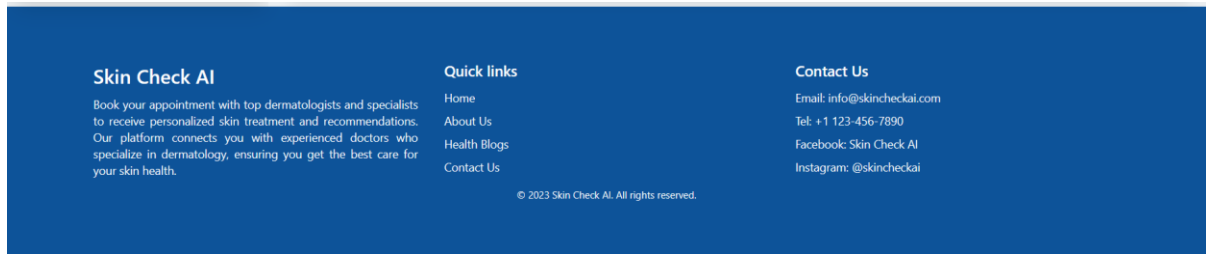


The "Messages" option provides patients with a platform to communicate directly with their doctors. Patients can engage in text-based chat or have video conversations, facilitating effective communication and eliminating the need for separate communication channels.

3.9.4.6 Billing

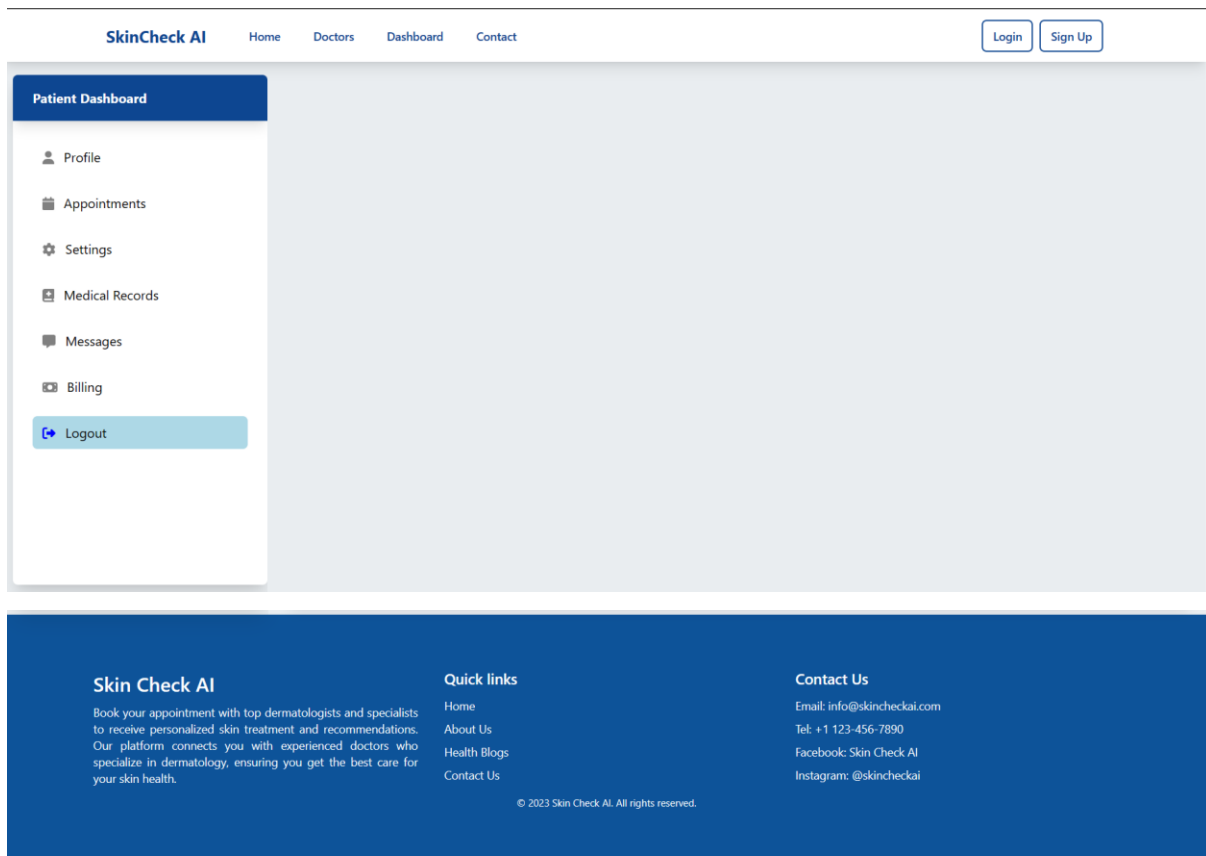


Requirements and Design



The "Billing" option allows patients to access their payment history. By selecting this option, patients can view the payments they have made, providing transparency and ensuring a clear overview of financial transactions related to their healthcare services.

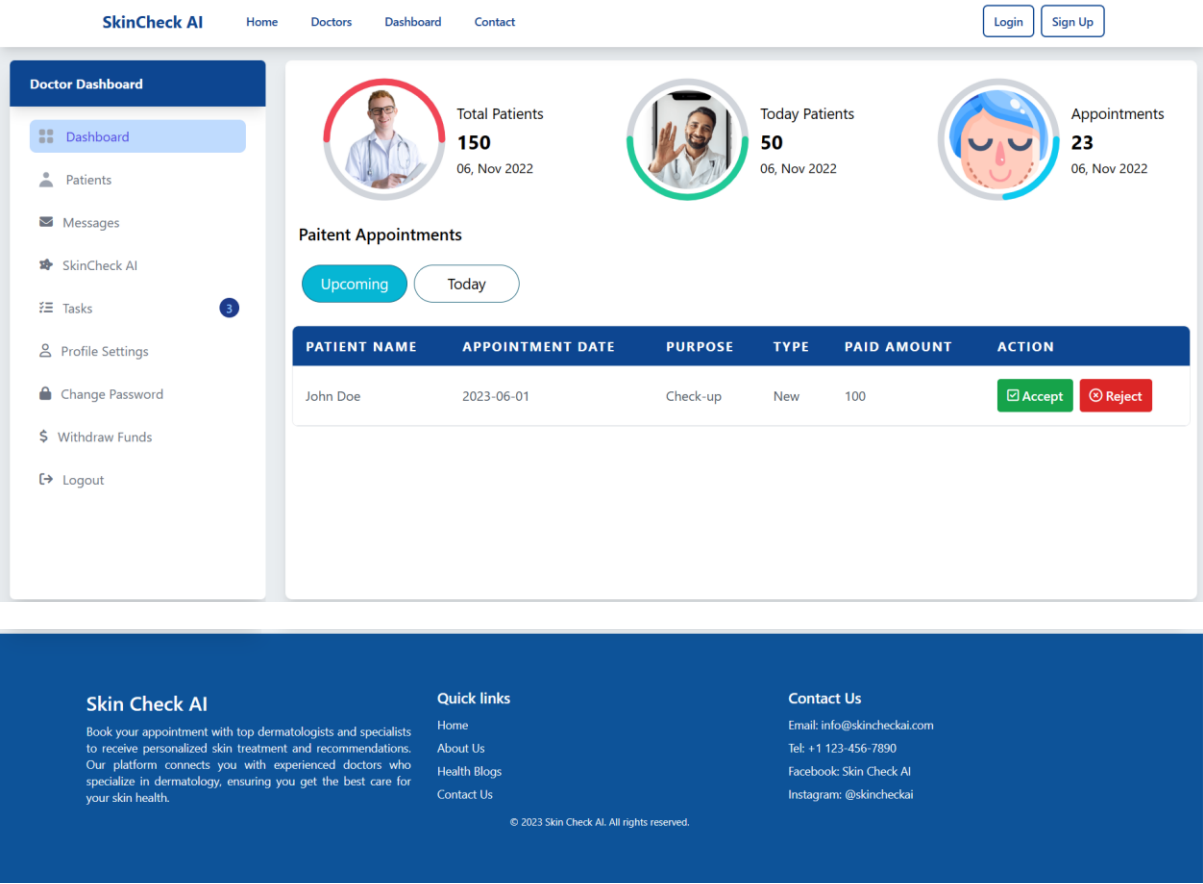
3.9.4.7 Logout



Finally, the "Logout" option offers patients a convenient way to securely log out of their accounts, ensuring privacy and account protection.

3.9.5 Doctors Dashboard

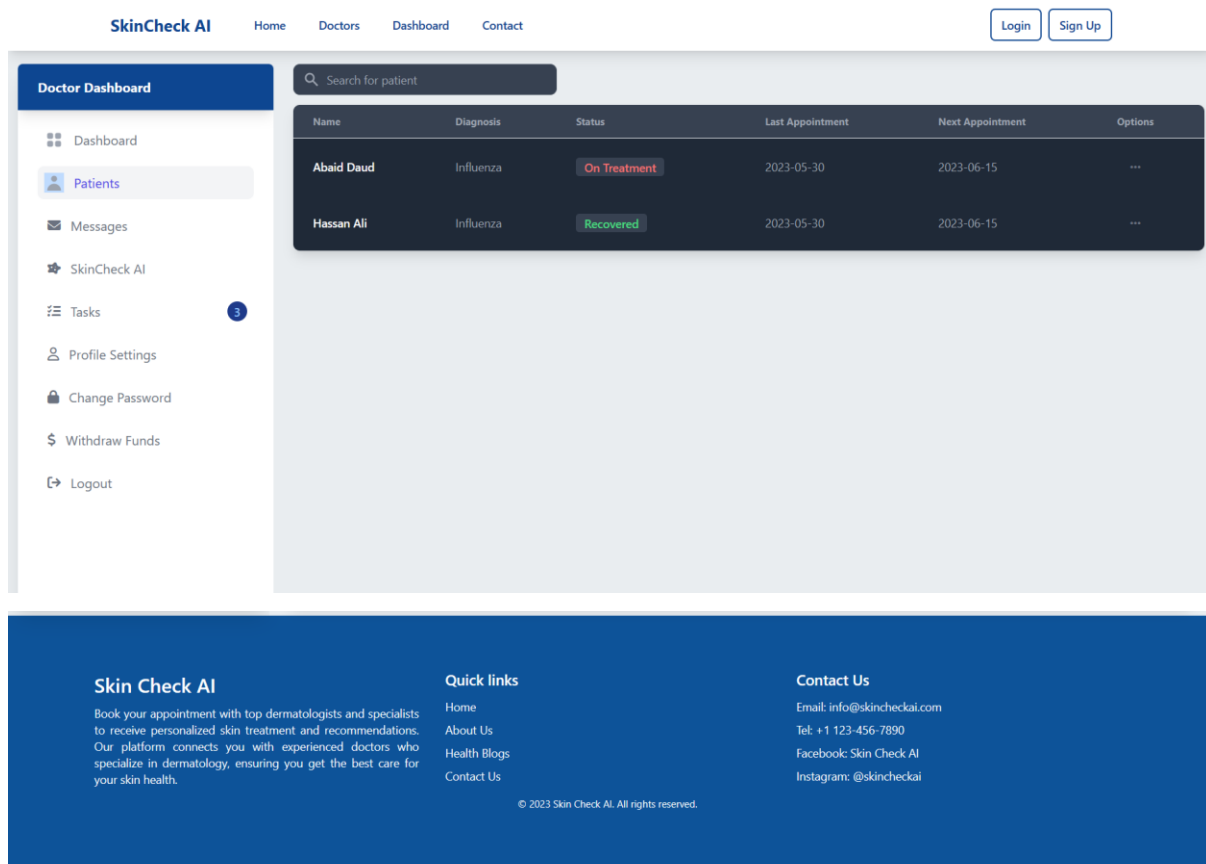
3.9.5.1 Dashboard



Upon accessing the doctor's dashboard, doctors can navigate to the "Dashboard" option in the navigation bar. This section provides an overview of their practice, displaying the total number of patients under their care, the number of patients scheduled for the current day, and a list of upcoming appointments. This information allows doctors to have a quick snapshot of their patient load and daily schedule. Additionally, doctors can accept or reject upcoming appointments based on their availability and preferences, ensuring optimal time management.

Requirements and Design

3.9.5.2 Patients

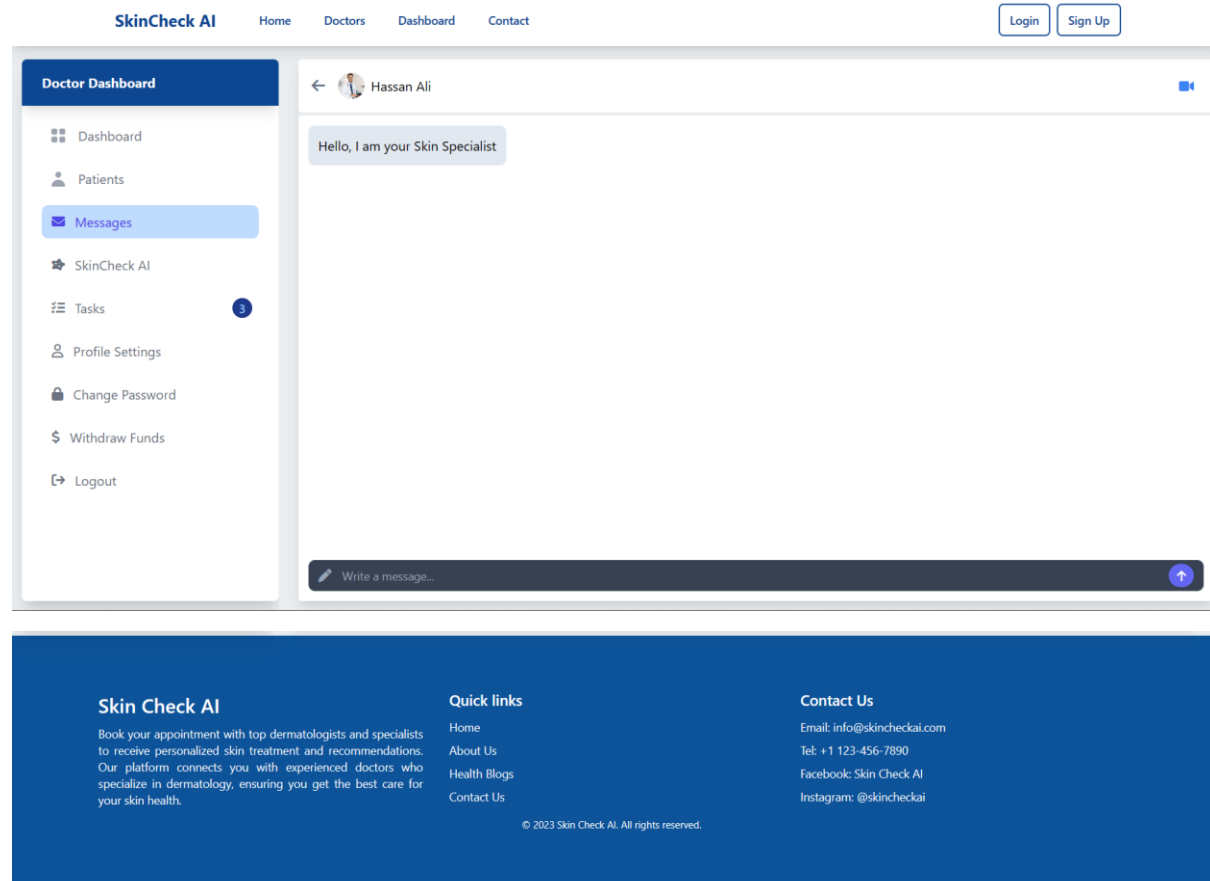


The "Patients" option in the navigation bar allows doctors to access detailed patient information. Doctors can view patient details such as their name, diagnosis, appointment status, last appointment date, and the next appointment. This feature facilitates easy access to patient records and helps doctors keep track of their patients' medical history. Furthermore, doctors have the ability to search for specific patients by name, enabling efficient retrieval of patient information.

3.9.5.3 Messages

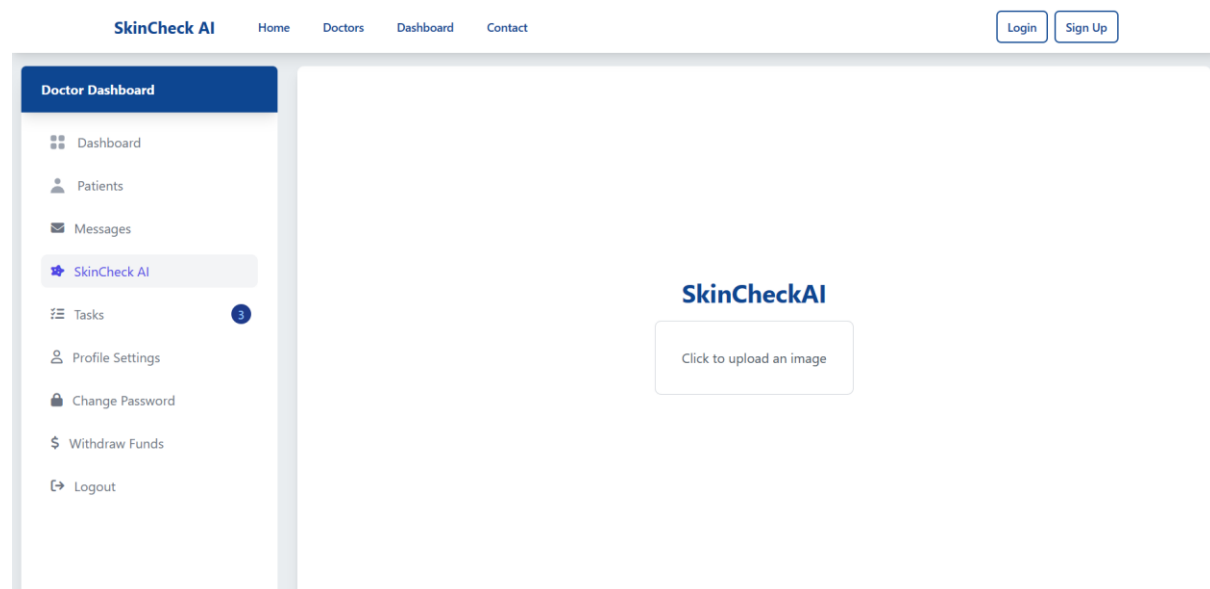
SkinCheck AI

3 0 5 4 Messages

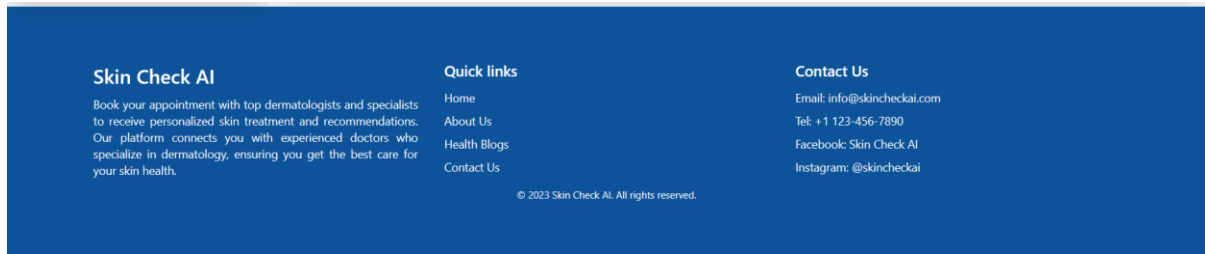


The "Messages" option provides patients with a platform to communicate directly with their patients. Doctors can engage in text-based chat or have video conversations, facilitating effective communication and eliminating the need for separate communication channels.

3.8.5.4 SkinCheck AI

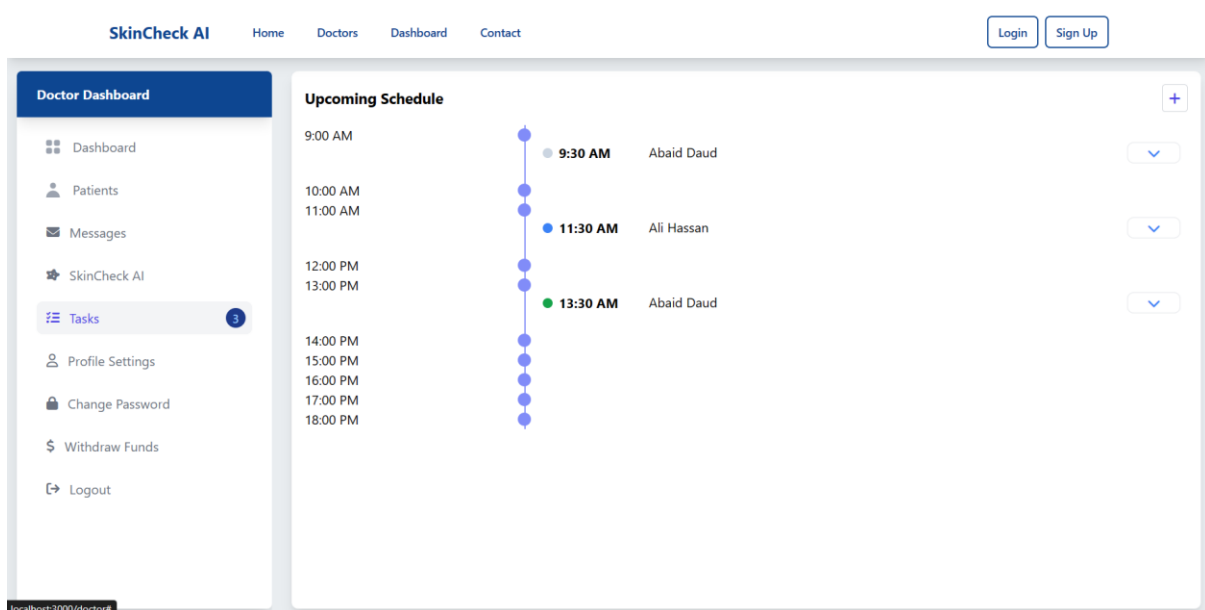


Requirements and Design



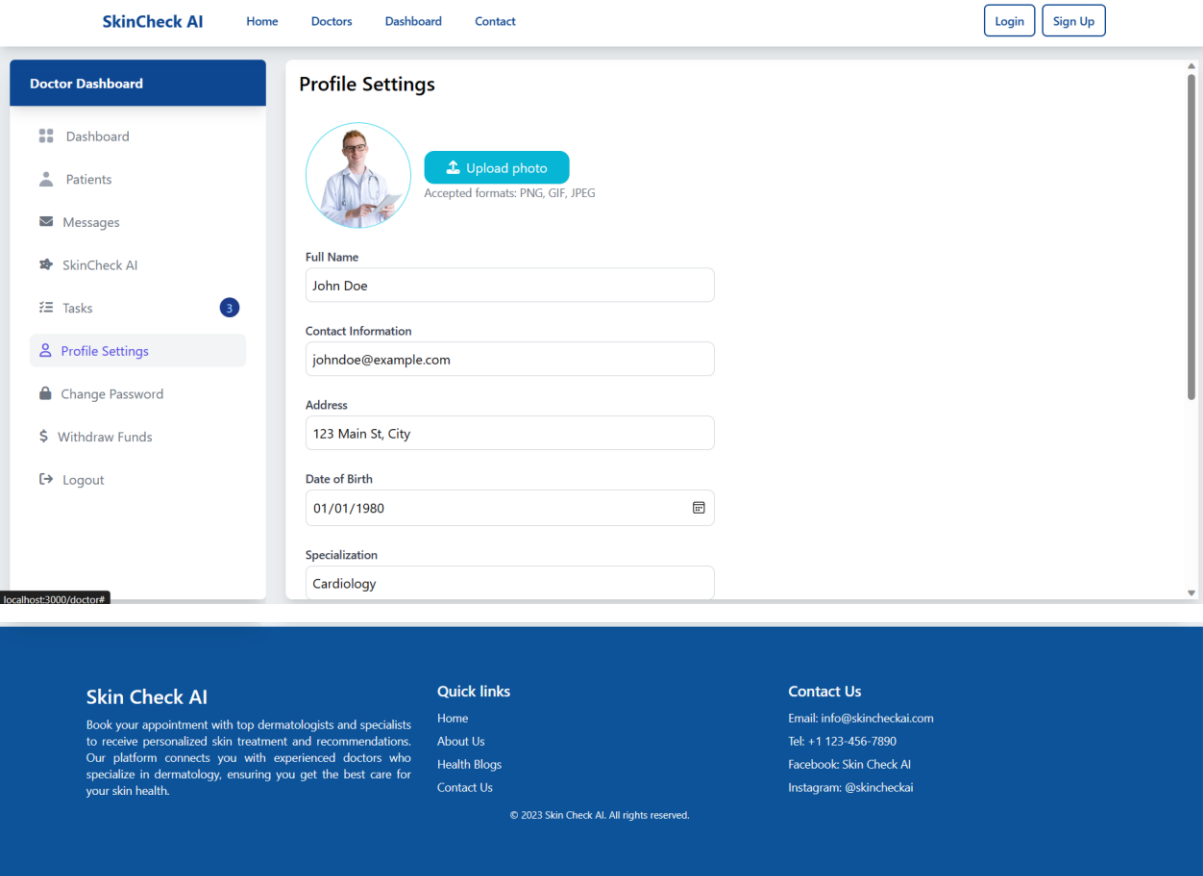
The "SkinCheck AI" option in the navigation bar provides doctors with a valuable tool for second opinions on skin diseases. Doctors can upload an image of a skin disease, and the AI-powered system will analyze the image and provide the name of the disease. This feature assists doctors in confirming diagnoses and expanding their knowledge base.

3.9.5.5 Tasks



The "Tasks" option in the doctor's dashboard offers doctors a convenient feature to manage their tasks and to-do lists effectively. This feature allows doctors to stay organized and ensure timely completion of important tasks related to patient care and administrative duties.

3.9.5.6 Profile Setting



The "Profile Settings" option enables doctors to update and manage their profile information. Doctors can change their name, email address, date of birth (DOB), specialization, qualifications, and add their experience and qualifications. This ensures that doctors can keep their profile up to date, reflecting their expertise and credentials accurately.

3.9.5.7 Change Password

The screenshot displays the 'Doctor Dashboard' for 'SkinCheck AI'. The dashboard includes a sidebar with navigation options: Dashboard, Patients, Messages, SkinCheck AI, Tasks (with a notification badge), Profile Settings, Change Password (highlighted), Withdraw Funds, and Logout. The main content area is titled 'Change Password' and contains three input fields for 'Current Password', 'New Password', and 'Confirm New Password', each with a masked password display (*****). A 'Save Password' button is located below the input fields. The footer section is dark blue and contains three columns of information: 'Skin Check AI' with a description of the service, 'Quick links' with links to Home, About Us, Health Blogs, and Contact Us, and 'Contact Us' with email, phone, and social media details. A copyright notice '© 2023 Skin Check AI. All rights reserved.' is centered at the bottom.

SkinCheck AI Home Doctors Dashboard Contact

Login Sign Up

Doctor Dashboard

- Dashboard
- Patients
- Messages
- SkinCheck AI
- Tasks 3
- Profile Settings
- Change Password**
- Withdraw Funds
- Logout

Change Password

Current Password

New Password

Confirm New Password

Save Password

Skin Check AI
Book your appointment with top dermatologists and specialists to receive personalized skin treatment and recommendations. Our platform connects you with experienced doctors who specialize in dermatology, ensuring you get the best care for your skin health.

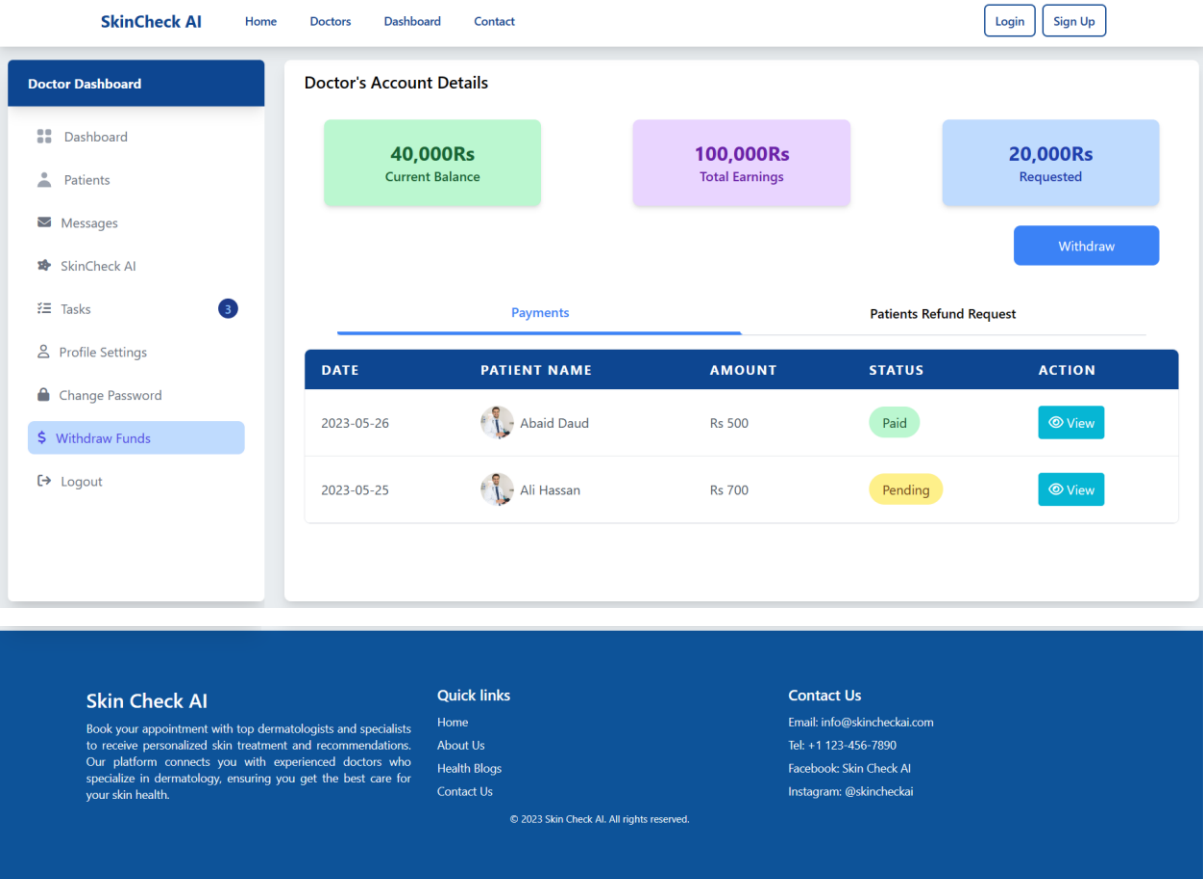
Quick links
Home
About Us
Health Blogs
Contact Us

Contact Us
Email: info@skincheckai.com
Tel: +1 123-456-7890
Facebook: Skin Check AI
Instagram: @skincheckai

© 2023 Skin Check AI. All rights reserved.

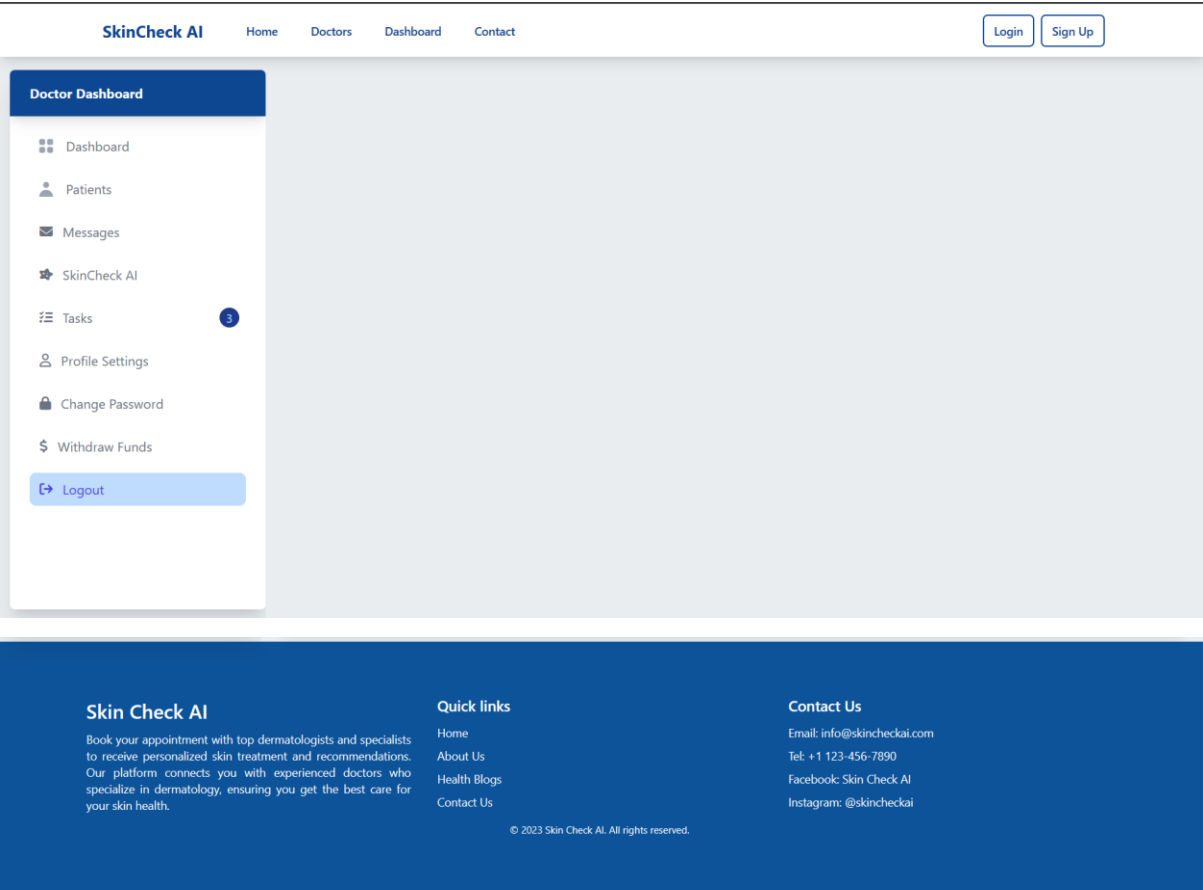
The "Change Password" option allows doctors to update their account password securely. This feature ensures that doctors can maintain the security of their account and access to confidential patient information.

3.9.5.8 Withdraw Funds



The "Withdraw Funds" option enables doctors to review payment records and withdraw their earned payments. Doctors can also track any refund requests made by patients, ensuring transparency and efficient financial management.

3.9.5.9 Logout



Finally, the "Logout" option offers doctors a convenient way to securely log out of their accounts, ensuring privacy and account protection.

Chapter 4: Implementation and Test Cases

For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions.

4.1 Implementation

4.1.1 Implementation of First Component/Algorithm

The first component of the system involves building and training a deep learning model using the ResNet50 architecture, implemented using the Keras library. The implementation steps are as follows:

1. **Importing the necessary libraries and modules:** The required libraries such as drive, os, image, cv2, numpy, pandas, matplotlib, tensorflow, keras, and pickle are imported to provide functionalities for accessing the dataset, preprocessing images, data manipulation, visualization, building and training the model, and saving/loading data.
2. **Preprocessing the dataset:** The dataset is prepared for training by performing the following steps:
 - The directory path to the dataset is set.
 - Categories (skin disease labels) are defined.
 - The frequency of data in each category is counted and visualized using matplotlib.
 - A sample image from each category is loaded and resized for visualization purposes.
3. **Creating training and testing data:** Functions are defined to create the training and testing data. The steps involved are:
 - Looping over each category and reading and resizing the images using cv2.
 - Appending the resized images and their corresponding labels to the training and testing data lists.
 - Shuffling the training and testing data to ensure randomness.
4. **Preparing the data for training:** The data is prepared for training by performing the following steps:
 - Converting the training and testing data lists to numpy arrays for compatibility with the model.
 - Reshaping the images to match the input shape required by the model.
 - One-hot encoding the labels using to_categorical from tensorflow.keras.utils.
 - Splitting the data into training and testing sets.
5. **Building the model:** The model architecture is constructed as follows:

Implementation and Test Cases

- The input data is normalized by dividing it by 255.
 - The input shape is defined based on the shape of the training images.
 - The ResNet50 model without the top layers is created and initialized with random weights.
 - Additional layers, including batch normalization, dense layers with activation functions, and dropout layers, are added on top of the ResNet50 model.
 - The model is compiled with the Adamax optimizer and categorical cross-entropy loss function.
 - The model architecture is summarized.
6. **Training the model:** The model is trained using the training data by performing the following steps:
- The learning rate, number of epochs, batch size, and number of classes are defined.
 - A learning rate reduction callback is set up to adjust the learning rate during training.
 - The model is fitted to the training data and validated on the testing data.
 - The training and testing accuracy and loss are printed.
7. **Evaluating the model:** The model's performance is evaluated by performing the following steps:
- Predictions are made on the testing data.
 - Sample predictions are visualized.
 - The confusion matrix and classification report are computed to assess the model's performance.

The implementation is done in Python using the Keras library, which is a high-level neural networks API running on top of TensorFlow. The code is executed in a Google Colab environment, which provides access to GPU resources for faster training. The APIs used include Keras for building the model, TensorFlow for data manipulation and training, and scikit-learn for evaluating the model's performance.

This implementation focuses on training the model and evaluating its performance on the task of skin disease classification. The ResNet50 model, with additional layers, is used as the underlying architecture to capture meaningful features from the input images.

4.2 Test case Design and description

This section will be added in FYP-II. Summarize the common attributes of test cases. This may include input constraints that must be true for every input in the set of associated test cases,

any shared environmental needs, any shared special procedural requirements, and any shared case dependencies. The following scheme is recommended for describing test cases in detail.

4.2.1 Sample Test case No.1

<Software component Name>			
<Reference>			
Test Case ID:	<i>Reference Number</i>	Test Date:	<i>Date</i>
Test case Version:	<i>Version number</i>	Use Case Reference(s):	<i>Relation to use cases</i>
Revision History:	<i>Refer to previous test case identity (if any)</i>		
Objective	<i>Need and scope of the testing</i>		
Product/Ver/Module:	<i>Refer to overall system being built and the place of this test case in it.</i>		
Environment:	<i>Necessary and desired properties of the test environment. (hardware/software)</i>		
Assumptions:	<i>Assumptions that might affect the testing process.</i>		
Pre-Requisite:	<i>Necessary condition that needs to be fulfilled prior to the test case.</i>		
Step No.	Execution description	Procedure result	
	<i>Events being tested.</i>	<i>Mention software response.</i>	
Comments:			
☐ Passed ☐ Failed ☐ Not Executed			

4.2.2 Sample Test case No.2

.

.

.

4.3 Test Metrics

Summarize here the common ground of attributes of test case metrics.

4.3.1 Sample Test case Matric.No.1

Metric:	Purpose
Number of Test Cases:	Total number of test cases that you have developed for your system.
Number of Test Cases Passed:	The number of test cases that successfully passed
Number of Test Cases Failed:	The number of test cases that failed
Test Case Defect Density:	(No of test cases failed * 100) No of test cases executed
Test Case Effectiveness:	No of defects detected using test cases *100 Total number of defects detected
Traceability Matrix:	Traceability is the ability to determine that each feature has a source in requirements and each requirement has a corresponding implemented feature.

4.3.2 Sample Test case Metric.No.2

Chapter 5: Experimental Results and Analysis

This chapter will be added in FYP-II. Give proper analysis and discussion of experimental results (in plain English text) along with tables of results. **For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions.**

Chapter 6: Conclusion and Future Directions

6.1 Introduction

In this chapter, we present the conclusion and summary of the work done in FYP-Part 1 of our Skin Disease Detection Website project. We discuss the findings and results achieved, evaluate whether the project scope was fully covered, and assess the fulfillment of project objectives. Furthermore, we address the challenges encountered during the implementation process and identify any remaining tasks that were not completed in FYP-Part 1. Lastly, we provide recommendations for future enhancements and improvements to the project, along with a proposed plan of work for FYP-Part 2.

6.2 Conclusions and Summary of Work Done

In FYP-Part 1, we have made significant progress in developing the Skin Disease Detection Website. We have successfully completed the front-end implementation of the website using React JS. The website includes various modules and features designed to cater to both patients and doctors. These features encompass functionalities such as doctor search, video consultations, appointment management, patient records, prescription management, messaging, billing, and more.

On the algorithmic front, we have trained a ResNet-50 model on a dataset consisting of five classes of skin diseases. Although limited to a train-test split, we have demonstrated the potential for accurate skin disease detection using deep learning techniques. We plan to further improve the algorithm's performance and accuracy by employing data augmentation techniques, normalization, segmentation, and the use of a threshold for disease prediction in FYP-Part 2.

6.3 Coverage of Project Scope and Objectives

Regarding the project scope, we have achieved 45% completion of the overall implementation, focusing primarily on the website front-end development. The patient and doctor dashboards, appointment management, profile settings, and other key features have been successfully implemented. However, the back-end development and integration with the algorithm for skin disease detection remain pending and will be addressed in FYP-Part 2.

The project objectives, as outlined in the initial proposal, have been partially met. We have successfully created a user-friendly website interface that facilitates patient-doctor interactions

and provides access to educational resources related to skin diseases. However, the core objective of implementing the skin disease detection algorithm and integrating it with the website's backend is yet to be accomplished.

6.4 Challenges Faced

Throughout the development process, we encountered several challenges. The complexity of front-end design and implementation required careful attention to ensure a seamless user experience. Overcoming technical hurdles and ensuring compatibility across different devices and browsers were key challenges. Additionally, acquiring and preparing a comprehensive dataset for training the skin disease detection algorithm posed difficulties, along with the need for expertise in deep learning techniques.

6.5 Remaining Work and Future Directions

In FYP-Part 2, we will focus on completing the remaining 55% of the project. The primary tasks include the development of the website backend and integration with the skin disease detection algorithm. We will train the algorithm using an augmented dataset, normalize the data, perform segmentation, and apply a threshold for disease prediction. The integration of the algorithm's predictions with the website's features, such as patient records and doctor recommendations, will provide an enhanced user experience.

6.6 Recommendations for Future Enhancements

To further enhance and improve the Skin Disease Detection Website, we propose the following recommendations for future work:

1. **Extend the Algorithm's Capability:** Expand the algorithm to detect a wider range of skin diseases by incorporating additional disease classes into the training dataset. This will increase the system's diagnostic capabilities and make it more valuable for dermatologists and patients.
2. **Advanced Image Analysis:** Incorporate advanced image analysis techniques, such as computer vision and pattern recognition algorithms, to improve the accuracy and efficiency of skin disease detection. This may involve the detection of specific patterns or features within skin images to aid in diagnosis.
3. **Integration with Electronic Health Records (EHR):** Integrate the website with existing electronic health record systems to streamline data management and ensure secure

Conclusion and Future Directions

storage of patient records. This will enable doctors to access comprehensive patient information and provide more personalized care.

4. **Mobile Application Development:** Develop a mobile application version of the website to provide users with easy access to the platform on their smartphones. This will increase convenience and expand the reach of the skin disease detection services.

6.7 Proposed Plan for FYP-Part 2

In FYP-Part 2, the following tasks will be undertaken to complete the project:

1. **Back-end Development:** Develop the website's back-end infrastructure using suitable technologies and frameworks. This will involve implementing the server-side logic, database integration, user authentication, and data storage.
2. **Algorithm Integration:** Integrate the trained skin disease detection algorithm with the website's back-end. Establish a pipeline to receive skin images, process them through the algorithm, and provide disease predictions and recommendations to doctors and patients.
3. **Algorithm Improvement:** Improve the algorithm's performance by employing data augmentation techniques, normalization, segmentation, and the use of a threshold for disease prediction. Evaluate the algorithm's accuracy and fine-tune it as necessary.
4. **Testing and Validation:** Conduct extensive testing and validation of the website and algorithm functionalities to ensure reliability, accuracy, and usability. Address any identified issues or bugs to deliver a robust and user-friendly system.
5. **Documentation and Final Report:** Document all the project details, including the architectural design, implementation decisions, algorithm training methodologies, and user instructions. Prepare the final report summarizing the entire project, including conclusions, future recommendations, and the proposed plan for FYP-Part 2.

By following this plan, we aim to successfully complete the Skin Disease Detection Website project and deliver a comprehensive and effective solution for dermatological diagnosis and patient care.

References

- [1] R. H. H. Hridoy, F. Akter and A. Rakshit, "Computer Vision Based Skin Disorder Recognition using EfficientNet: A Transfer Learning Approach," in *Rashidul Hasan Hridoy*, Cairo, Egypt, 2021.
- [2] M. Wei, Q. Wu, H. Ji, J. Wang, T. Lyu, J. Liu and L. Zhao, "A Skin Disease Classification Model Based on DenseNet and ConvNeXt Fusion," *Electronics*, vol. 12, no. 2, pp. 1-19, 2023.
- [3] B. Zhang, X. Zhou, Y. Luo, H. Zhang, H. Yang, J. Ma and L. Ma, "Opportunities and Challenges: Classification of Skin Disease Based on Deep Learning," *Chinese Journal of Mechanical Engineering*, vol. 34, no. 5, pp. 1-14, 2021.
- [4] V. Anand, S. Gupta, D. Koundal, S. Mahajan, A. K. Pandit and A. Zaguia, "Deep Learning Based Automated Diagnosis of Skin Diseases Using Dermoscopy," *Computers, Materials & Continua*, vol. 71, no. 5, pp. 1-16, 2022.